

BIO 3105:
Biology for Engineers
Chapter-07: Biotechnology

Lecture-13,14,15

17/09/2025

21/09/2025

24/09/2025

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Contents of the Chapter:

- **Biotechnology: Principles and Processes**
- **Processes of Recombinant DNA Technology**
- **Tools of Biotechnology**
- **Biotechnology Applications In Medicine**
- **Biotechnology Applications In Agriculture**
- **Transgenic Animals and Ethical Issues**

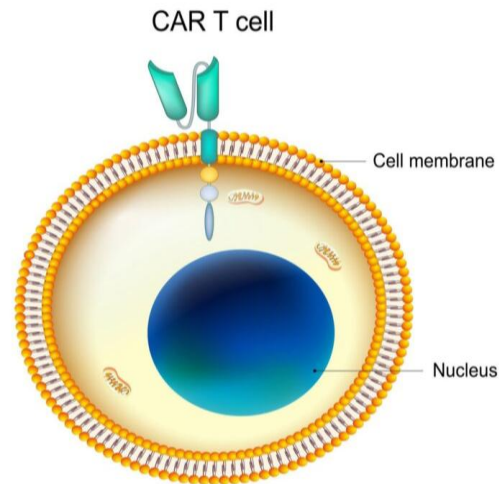
Some Application of Biotechnology

- Medicine
- Research
- Industrial Biotechnology
- Agriculture

mRNA Vaccine



CAR-T cells to treat
Blood Cancers



rh Insulin



Wheat blast outbreak
and management



Biotechnology at a glance:

Biotechnology utilizes living cells and cellular materials to **create pharmaceutical, diagnostic, agricultural, environmental, and other products to benefit society**. It is also used to study and to alter genetic information in animals so that human diseases can be modeled and studied.

Like earlier technologies, biotechnology promises to provide many advantages, especially in three broad categories: **environmental protection, higher yields, and improvements in human health**.

Is Biotechnology Safe? **Biotechnology can be safe, but potential risks exist.** The continued safe use of biotechnology requires constant vigilance on a case-by-case basis. Scientists are responsible for careful preparation of recombinant organisms involved in research or product development.

What is the future of biotechnology?

Automation and data-driven processes are increasingly being incorporated into biotechnology, and **anticipated to radically improve the predictability and reproducibility of research and manufacturing outcomes by 2040**.

Today, biotechnology is being used in countless areas including **agriculture, clinical applications, bioremediation and forensics**, where DNA sequencing is a common practice. Industry and medicine alike use the techniques of PCR, immunoassays and recombinant DNA.

Commonly used Techniques in biotechnology

1. **Polymerase chain reaction (PCR):** is a method widely used to make millions to billions of copies of a specific DNA sample rapidly,
2. **Gel electrophoresis:** is important to biotechnologists because it allows them to analyze the structure and function of macromolecules, such as DNA, RNA, etc.
3. **Recombinant DNA technology:** is useful in knowing defective genes in the fetuses. Some of these genes can also be repaired. (iv) Gene Library: The various ...**Genetic Engineering:** Involves direct manipulation of the **genetic** material of a living thing, for example, creating a bacteria to produce human insulin and ...
4. **DNA/RNA sequencing:** and manipulation of **DNA** are sometimes referred to as **DNA** technology. For example, for the cystic fibrosis gene therapy trial, ...
5. **Restriction Enzyme.** An **enzyme** from bacteria that can recognize specific base sequences in DNA and cut the DNA at that site. A **restriction enzyme** acts as a scissors.
6. **Molecular cloning:** Molecular **cloning** is crucial for biotechnologists because it allows them to isolate, manipulate, and study specific genes or DNA .

Commonly used Techniques in biotechnology

7. **Southern blotting:** involves the transfer of DNA to a nylon membrane, while northern **blotting** is the transfer of RNA to a nylon membrane.
8. **Chromatography:** In **chromatography**, components to be separated are distributed between two phases:- the stationary phase and the mobile phase.
9. **Reporter Gene for Gene Monitoring:** One of the technical problems encountered in attempts at **gene** transfer is knowing whether a particular **gene** has actually Includes a number of different technologies including Recombinant DNA (**GMO**) and Gene Editing. Can be used to insert genes or knock them out, either randomly or ...
10. **Spectrophotometer:** Every compound absorbs and emits lights over a certain range of wavelength. A **spectrophotometer** is an instrument that emits light of ...
11. **Western blot:** western blots are used to analyze the presence and expression of specific proteins in a variety of samples, such as cells, tissues, and body ...
12. **Enzyme-linked immunoassay (ELISA):** ELISA is used to detect and quantify specific proteins or antigens in a variety of samples, such as cells, tissues, and ...

DNA complementary base pairing

The two strands of a double strand DNA(dsDNA) molecule run anti-parallel to one another



5' T G A C C G T 3'
3' A C T G G C A 5'

The two strands of a DNA molecule are complementary - the sequence TGACCGT on one strand is complementary to ACTGGCA on the other strand. As the strands are anti-parallel these sequences are more precisely termed 5' TGACCGT 3' and 3' ACTGGCA 5'.

DNA base pairing allows the inheritance of nucleotide sequences

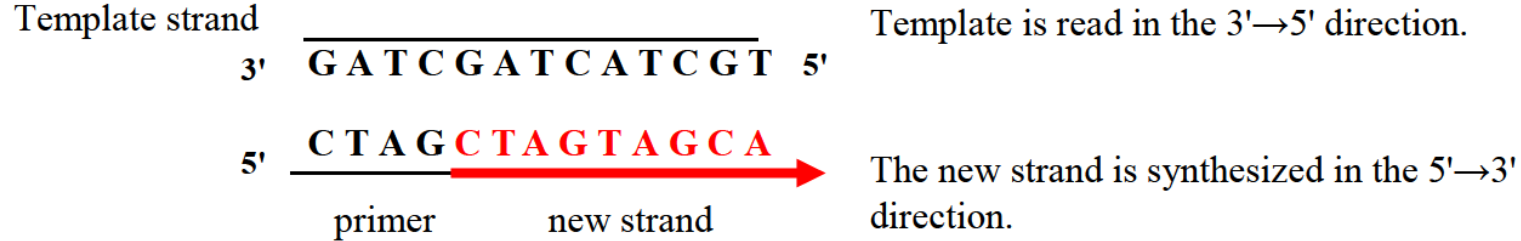
- As DNA contains important genetic information, it must therefore be able to replicate accurately.
- The specificity of interaction between the bases ensures that the sequence of nucleotides is preserved when the DNA replicates.
- During replication the two strands of the double stranded DNA (dsDNA) molecule separate.
- Each single strand acts as a template to synthesise a new strand.
- Due to complementary base pairing, the identity of the bases in the template strand defines the sequence of bases in the newly synthesised strand.
- This means that the sequence of the newly synthesised strand is complementary to that of the template.

DNA replication:

DNA replication requires:

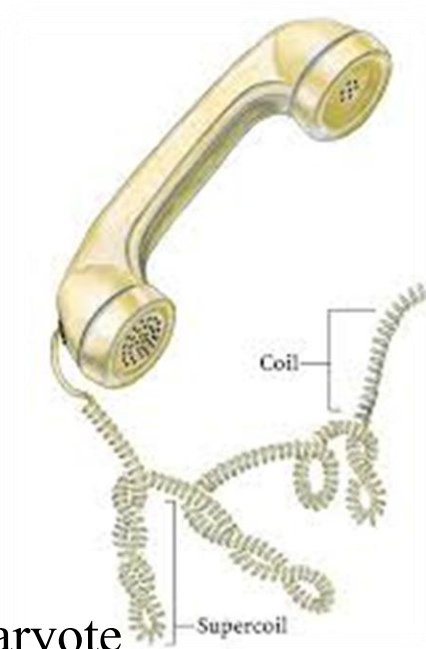
- A single strand of DNA to serve as **template** for the synthesis of a new dsDNA strand.
- A **primer**, a short oligonucleotide (a chain of up to 20 nucleotides) that is complementary in sequence to the DNA template and hence can bind the template by base pairing.
- The primer is essential to provide a free pre-existing 3'OH for attack on the 5'-phosphate of an incoming deoxyribonucleotide triphosphate (dNTP).
- All 4 dNTPs (dATP, dGTP, dCTP, dTTP).
- DNA polymerase (the enzyme that catalyses template-dependent DNA biosynthesis).

DNA replication...

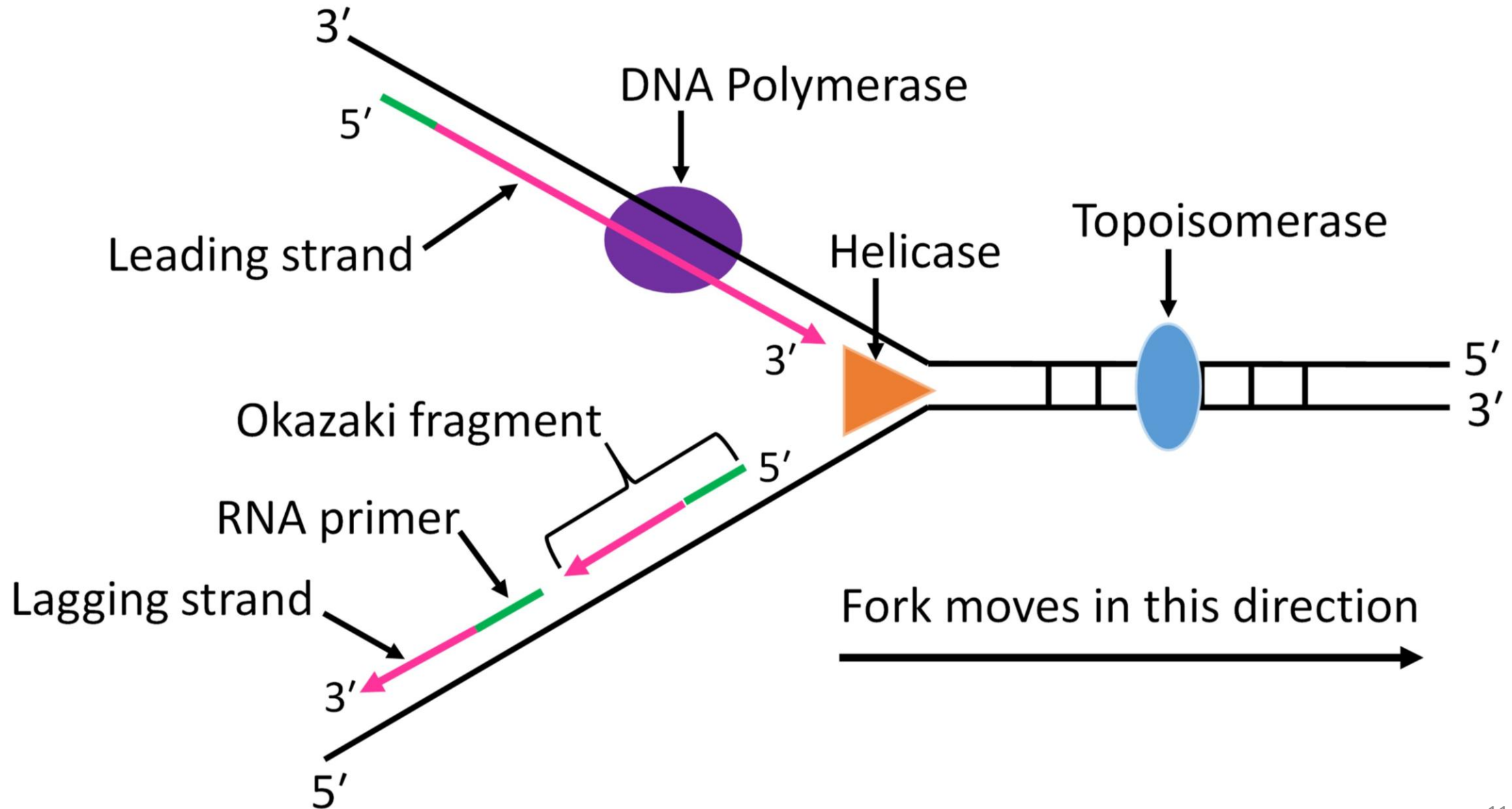


Proteins and Enzymes involved in DNA Replication:-

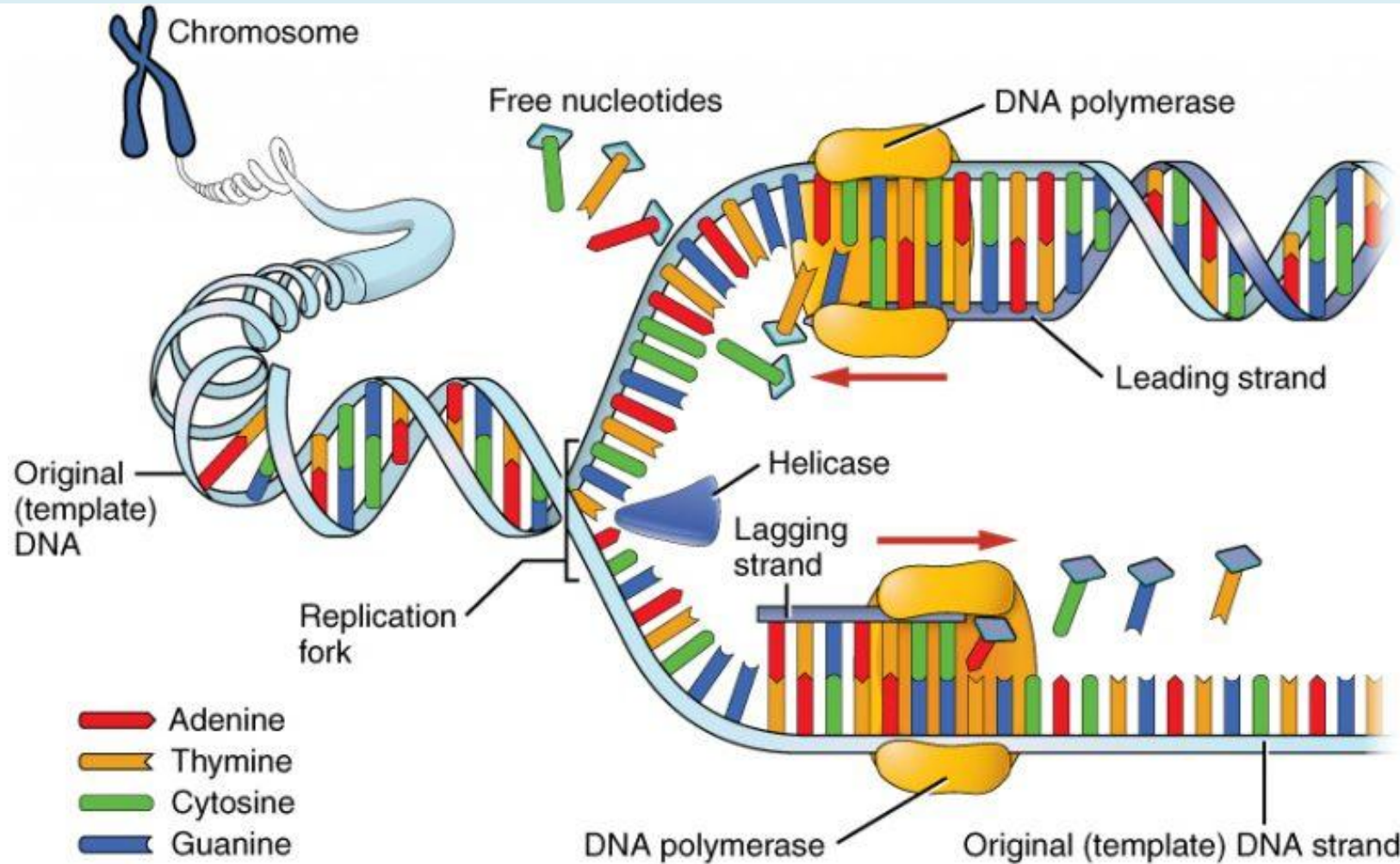
- **Topoisomerase:** Unwinds the DNA superhelix
- **Helicase:** Unwinds the double-stranded DNA
- **Primase:** Synthesizes short segment of RNA primer
- **DNA polymerases:**
 - Replace the RNA primer by DNA
 - Add nucleotides
 - Proofreading activity
- **DNA ligase:** Seals the nicks remains in the DNA backbone
- **Telomerase:** Allow the completion of DNA synthesis of eukaryote



DNA Replication: Simplified Version



DNA Replication:



DNA replication faithfully duplicates the entire genome of the cell. During DNA replication, a number of different enzymes work together to pull apart the two strands so each strand can be used as a template to synthesize new complementary strands. The two new daughter DNA molecules each contain one pre-existing strand and one newly synthesized strand. Thus, DNA replication is said to be “semiconservative.”

3 steps of DNA Replication:

Step 1: Initiation:.

The two complementary strands are separated, much like unzipping a zipper. Special enzymes, including **helicase**, untwist and separate the two strands of DNA.

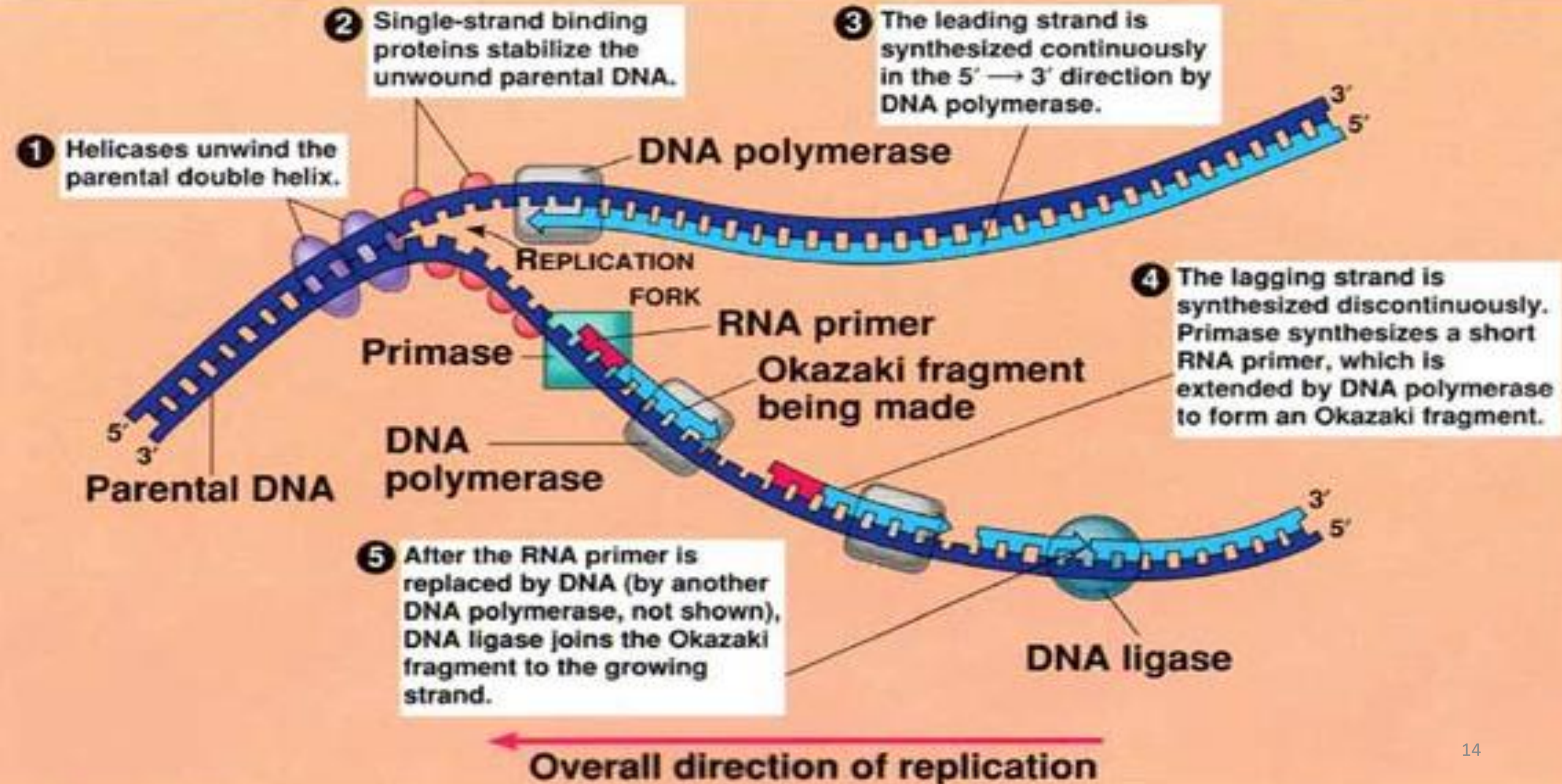
Step 2: Elongation:.

Each strand becomes a template along which a new complementary strand is built. DNA polymerase brings in the correct bases to complement the template strand, synthesizing a new strand base by base. A **DNA polymerase** is an enzyme that adds free nucleotides to the end of a chain of DNA, making a new double strand. This growing strand continues to be built until it has fully complemented the template strand.

Step 3: Termination:.

Once the two original strands are bound to their own, finished, complementary strands, DNA replication is stopped and the two new identical DNA molecules are complete.

DNA Replication:-Summary



Biotechnology: Principles and Processes

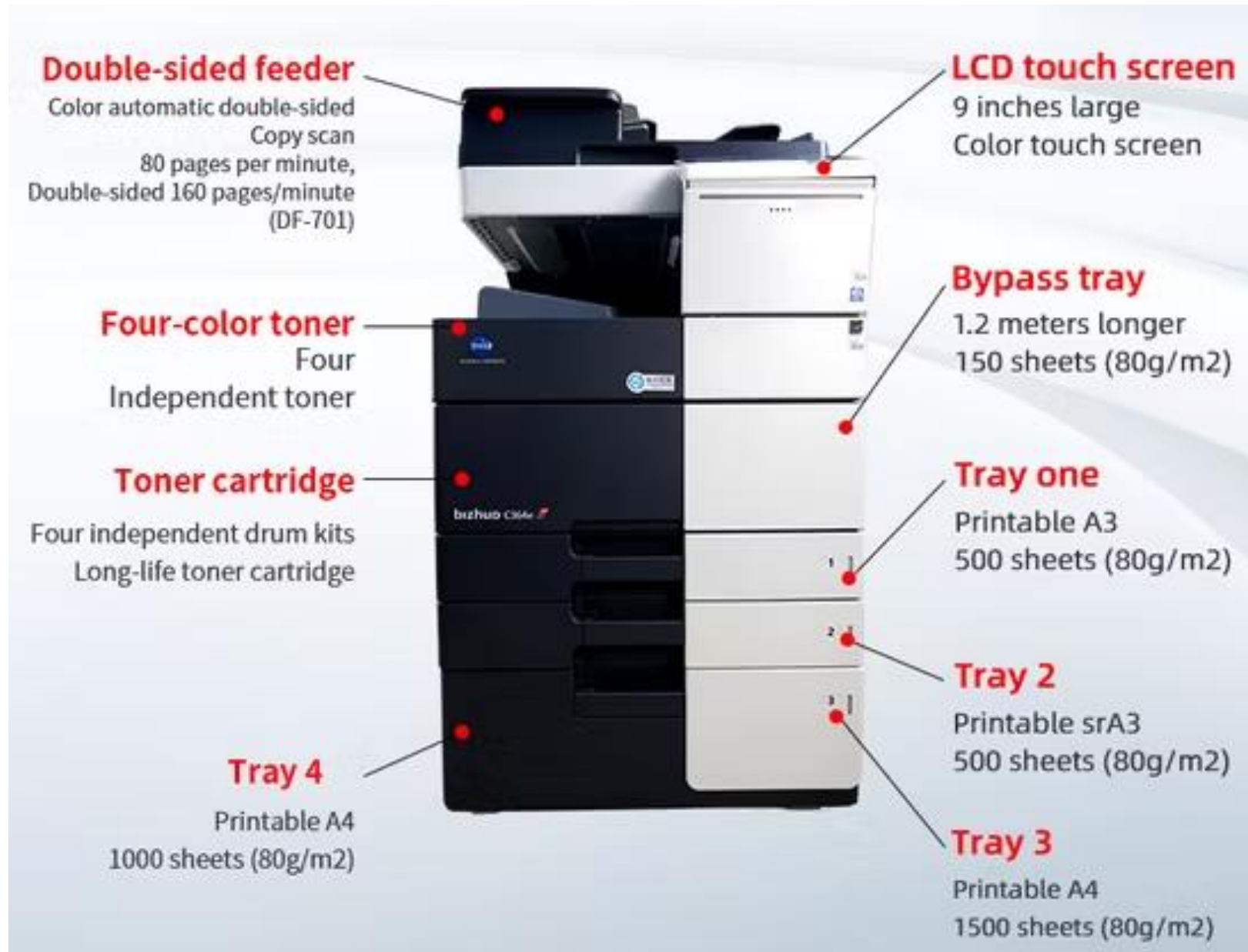
In biotechnology, scientists often need to amplify a small part of DNA or separate out different sized DNA fragments. How do scientists do this?

They use specific techniques such as

i) Polymerase Chain Reaction (PCR)

ii) Gel electrophoresis that allow them to achieve these goals. Let's learn about these techniques in detail.

A Photocopier/Photocopy Machine:



A Photocopier/Photocopy Machine to copy DNA/RNA



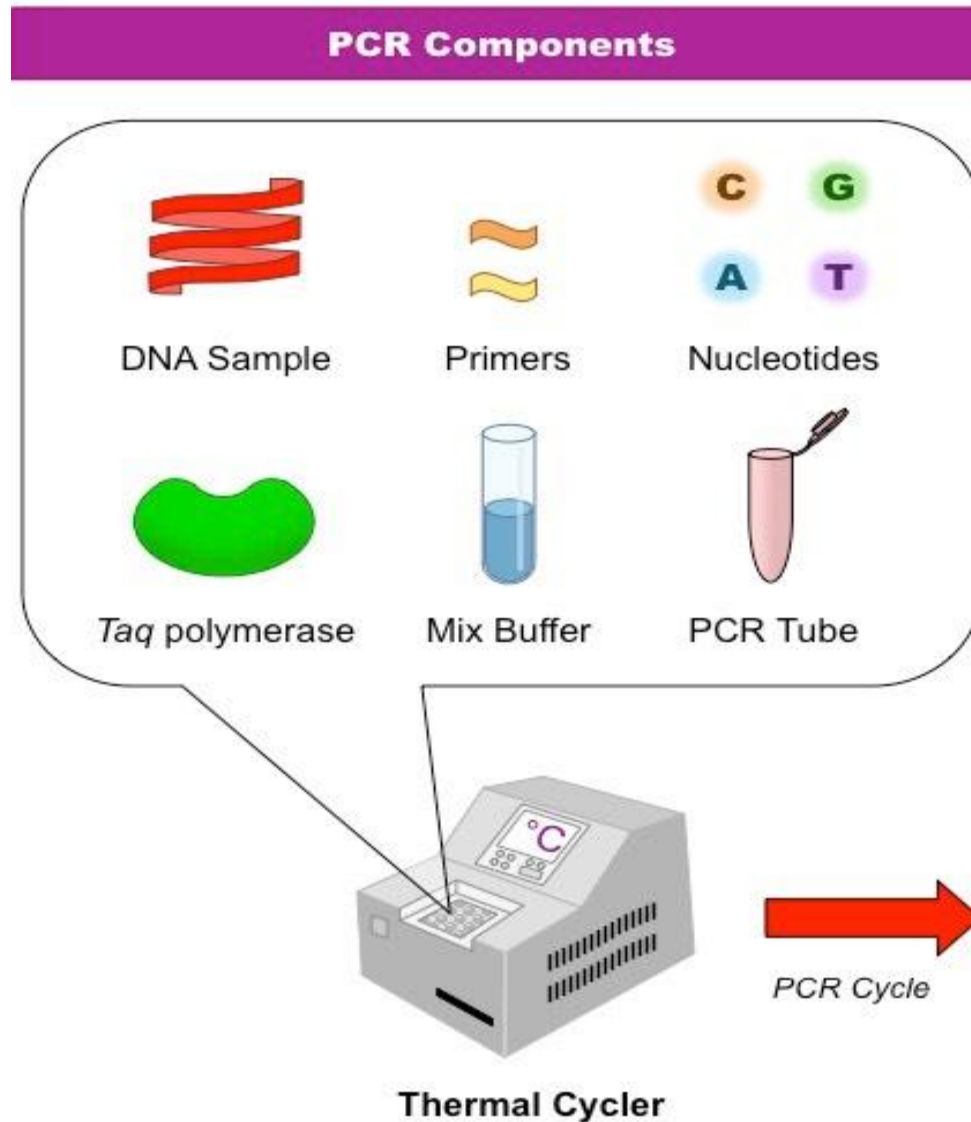
**A PCR Machine
Thermal Cycler**



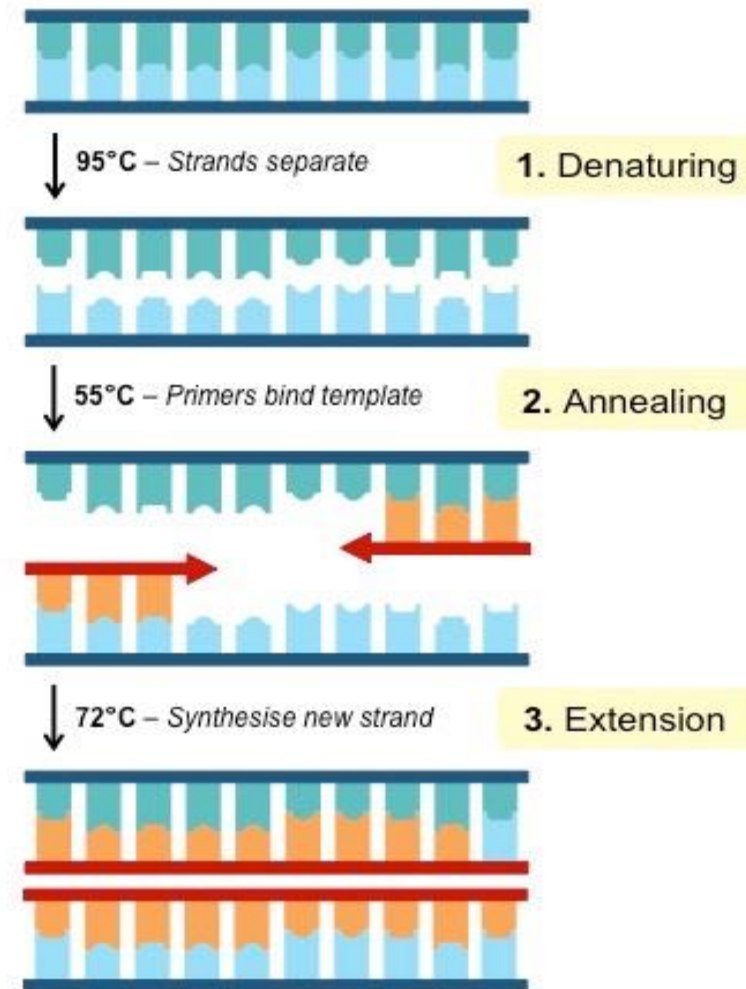
**A PCR Machine located at
UIU room 925**

PCR-Master mix and process:

PCR-Master mix



PCR Process (ONE Cycle)

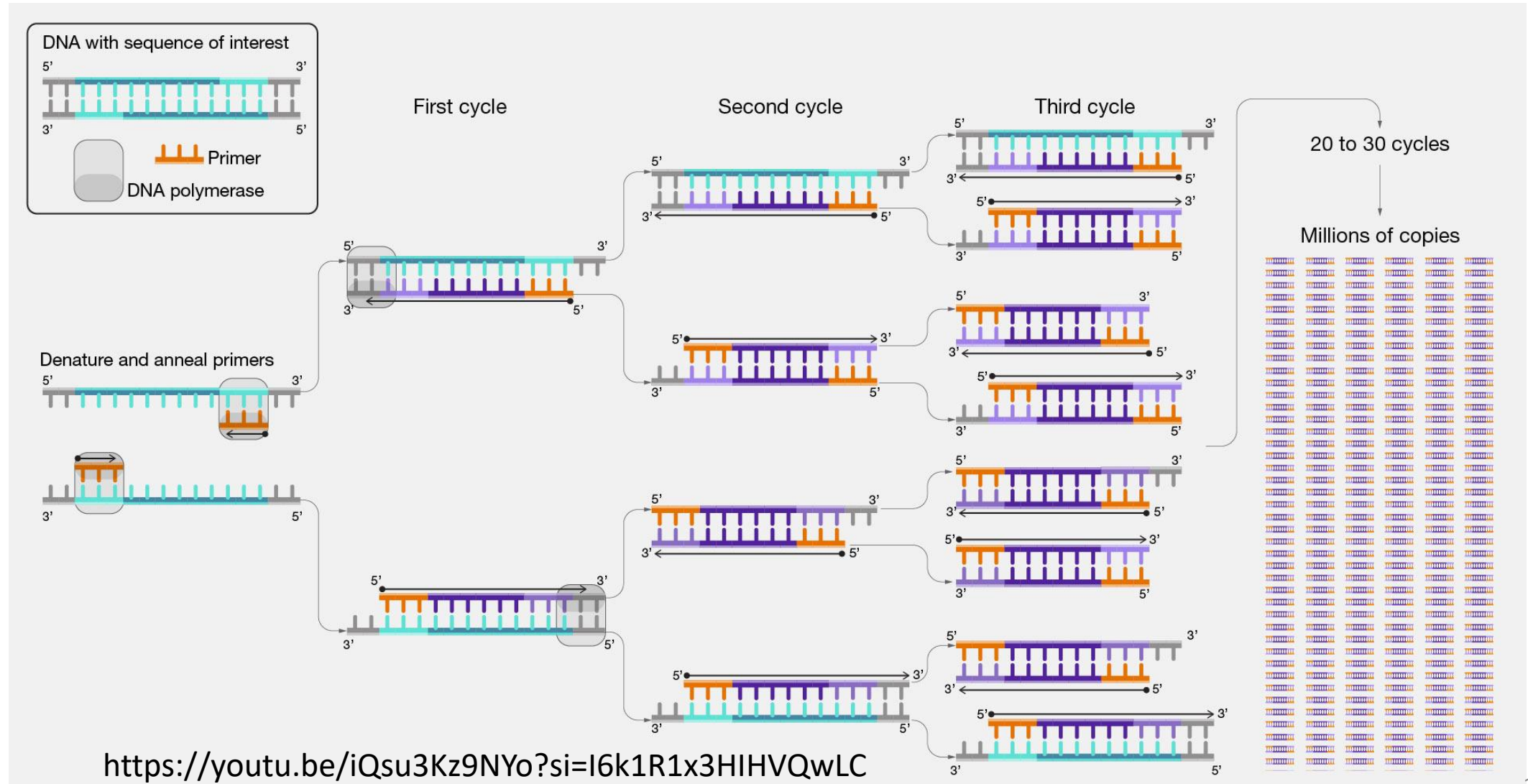


Polymerase Chain Reaction (PCR) components

- **Primers** – Short, chemically synthesized, single-stranded pieces of DNA that are complementary to the DNA fragment of interest.
- **DNA Polymerase** – The enzyme that elongates the primers by adding nucleotides to it, using the desired DNA fragment as a template. This enzyme needs to be able to withstand high temperatures used in the PCR reaction. Therefore, scientists isolated a thermostable DNA polymerase from the bacterium – *Thermus aquaticus*. It is known as Taq polymerase.
- **Nucleotides** – Single bases A, T, C, and G are the building blocks of DNA synthesis in a PCR reaction.
- **Buffer system** – The buffer contains potassium and magnesium that are essential for the DNA denaturation and renaturation steps. It also contains other factors important for enzyme activity, fidelity and stability.

Polymerase Chain Reaction (PCR)

Polymerase Chain Reaction (PCR) Polymerase chain reaction (abbreviated PCR) is a laboratory technique for rapidly producing (amplifying) millions to billions of copies of a specific segment of DNA, which can then be studied in greater detail.

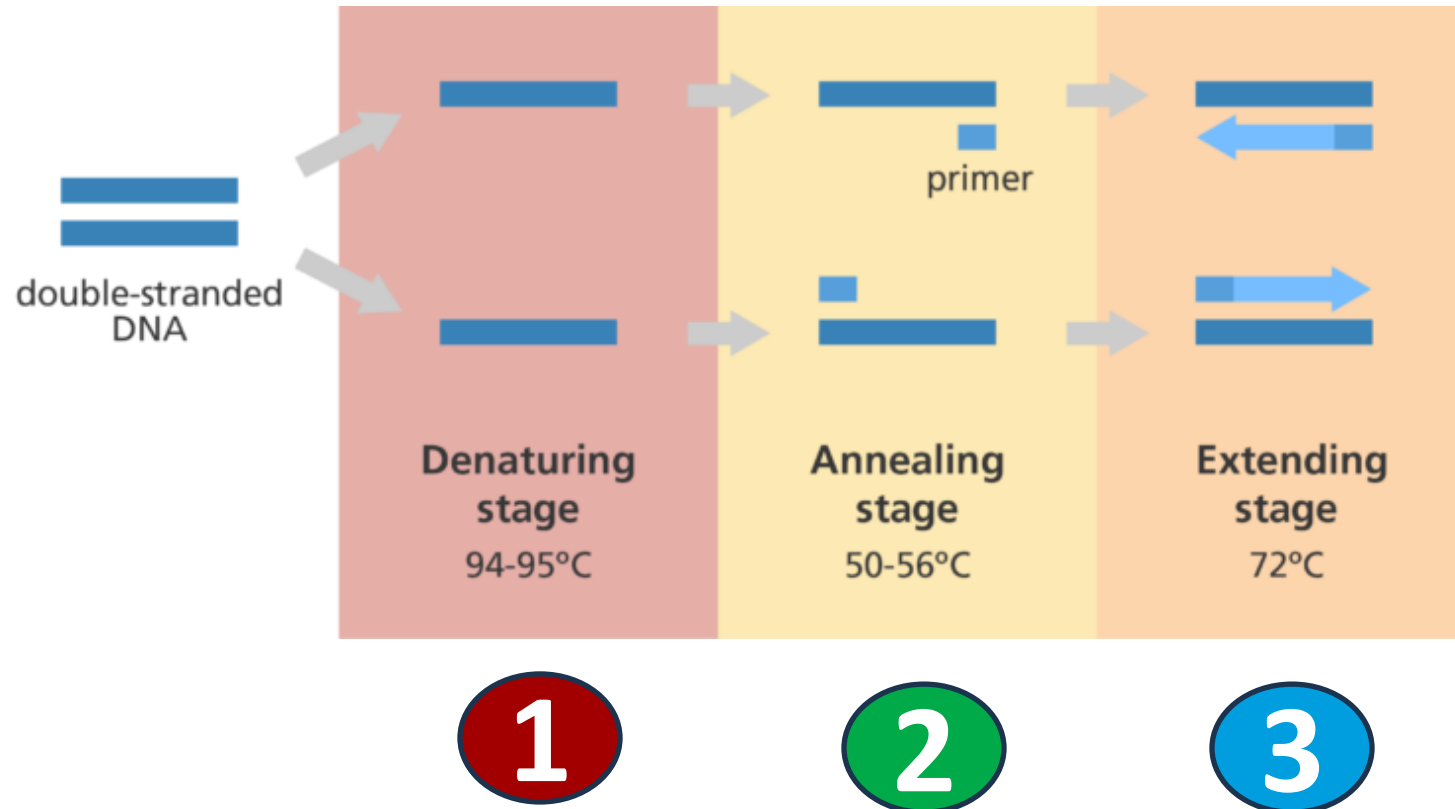


<https://youtu.be/iQsu3Kz9NYo?si=I6k1R1x3HIHVQwLC>

Polymerase Chain Reaction (PCR)

Polymerase Chain Reaction (PCR)

In a breakthrough discovery in 1985, Kary B. Mullis invented the Polymerase Chain Reaction (PCR) technique. This technique allows scientists to make millions of copies of a very small amount of DNA. It is based on the natural process of DNA replication that occurs within a cell. Let's learn about the components and steps of this technique. A PCR reaction needs the following key steps:



Steps in a PCR Reaction

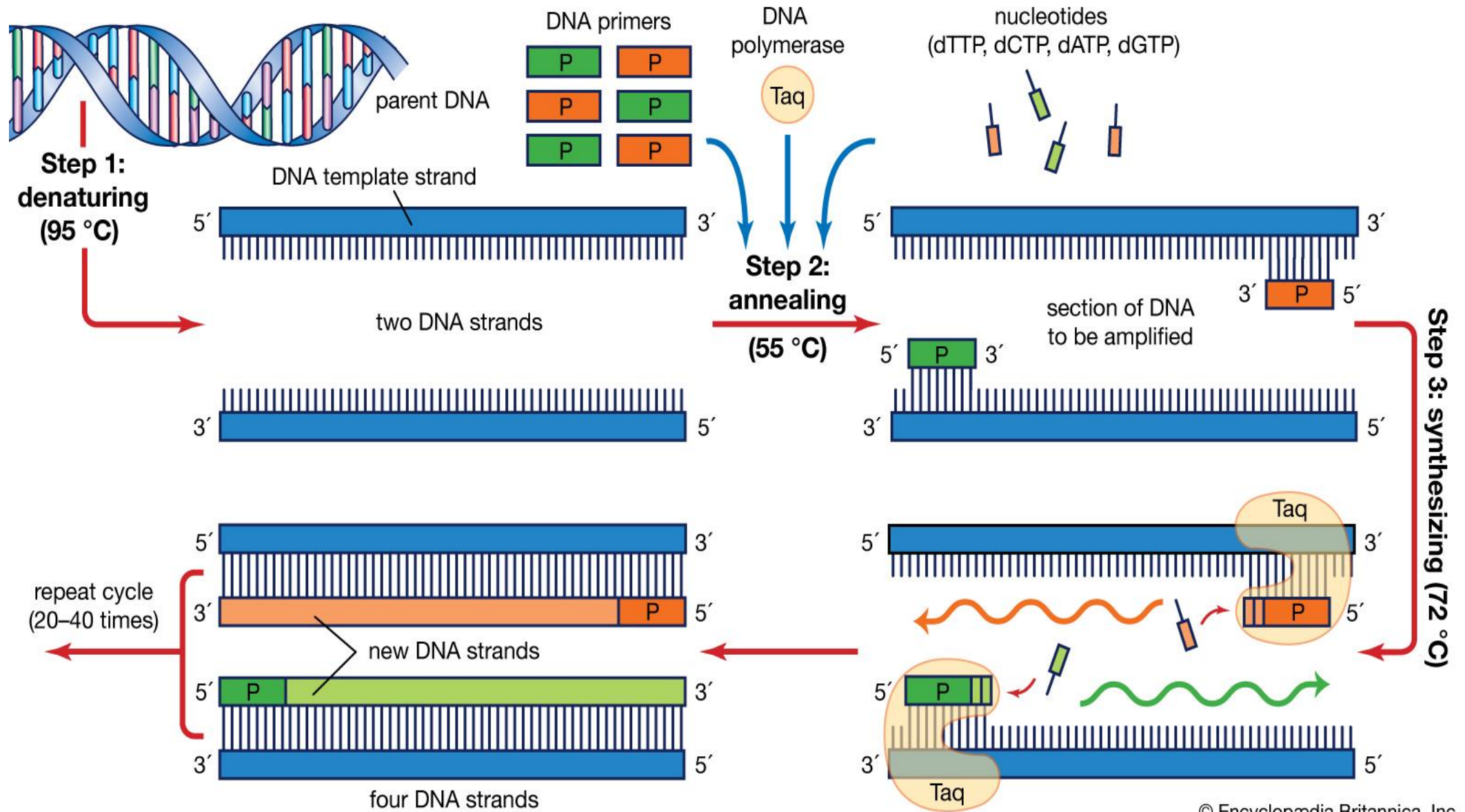
In a PCR reaction, DNA polymerase synthesizes a new DNA strand complementary to the DNA template by adding nucleotides to the primers. The above components are mixed together in PCR tubes and run on a thermal cycler, where they go through multiple cycles involving the following reactions:

- **Denaturation** – This step involves heating the reaction mixture to a high temperature (94°C). This denatures the double-stranded DNA template into single-strands by breaking the weak hydrogen bonds between the two DNA strands.
- **Annealing** – In this step, the reaction mixture is brought to a temperature of 54-60°C, to allow the primers to bind (anneal) to their complementary sequence in the template DNA.
- **Elongation/Extension** – In this step, the DNA polymerase sequentially adds nucleotides to the primers and extends it in the 5' to 3' direction.

One cycle of denaturation, annealing, and elongation amplifies the double-stranded DNA template to give two pieces of double-stranded DNA. The cycles keep repeating to produce more and more copies, increasing the number of copies exponentially.

Today, PCR is very important in many fields. In medicine, it is useful to detect genetic mutations and to detect HIV, malaria etc. in clinical samples.

Steps in a PCR Reaction



Additional readings

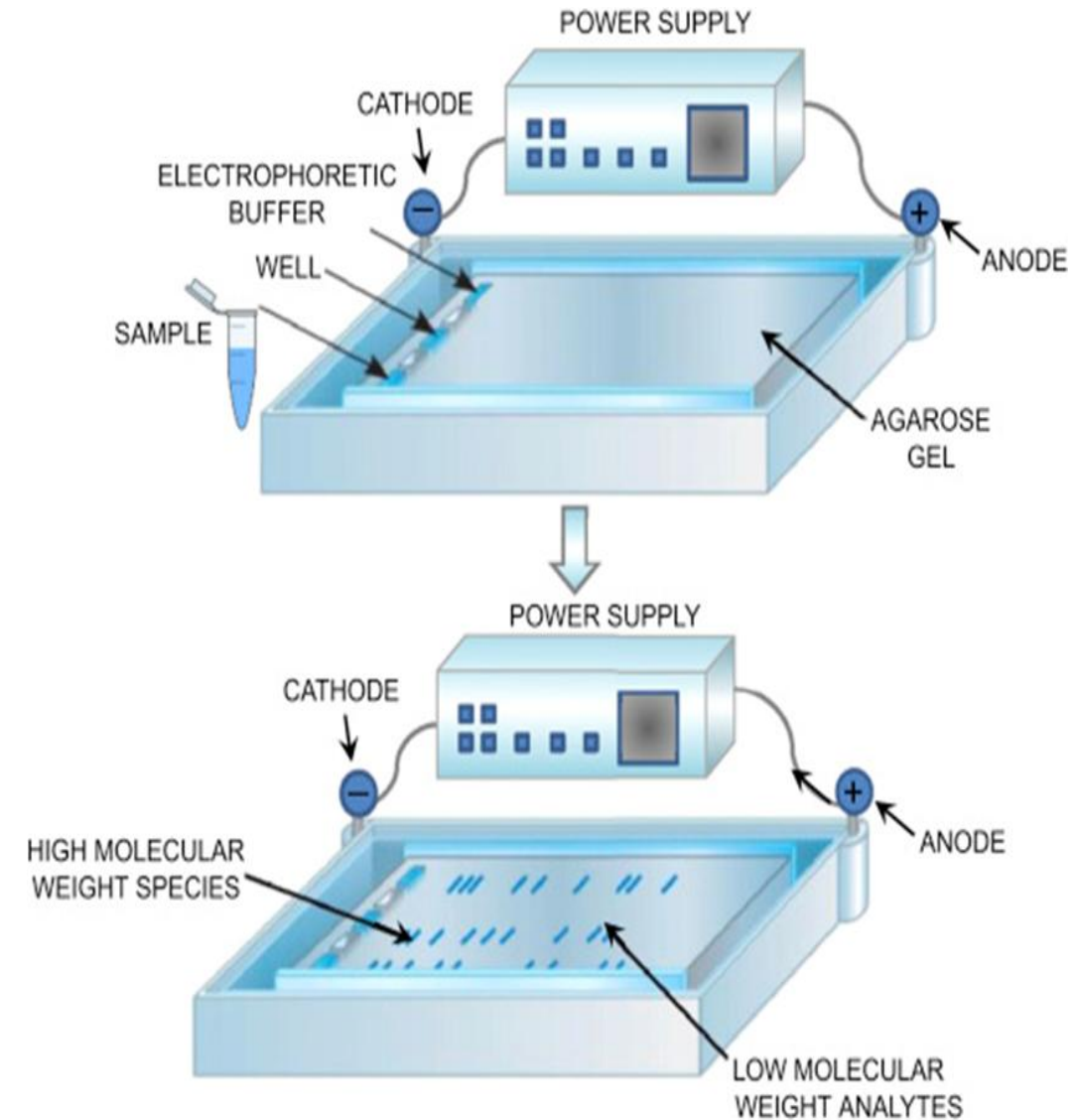
<https://praxilabs.com/en/blog/2021/04/28/what-are-the-three-basic-steps-of-conventional-pcr/>

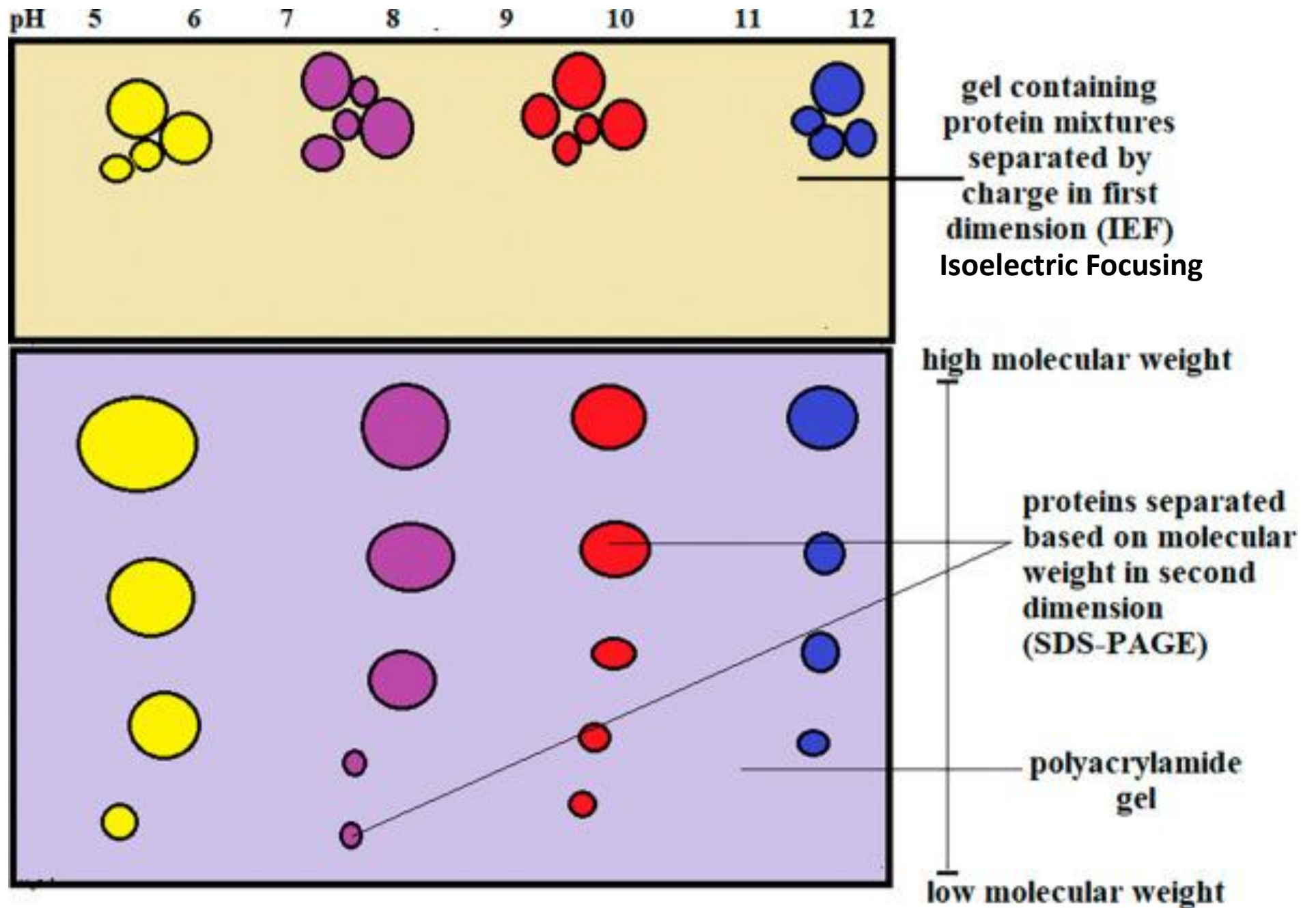
<https://www.aatbio.com/resources/application-notes/pcr-annealing-temperature-optimization>

<https://old-ib.bioninja.com.au/standard-level/topic-3-genetics/35-genetic-modification-and/pcr.html>

Gel Electrophoresis

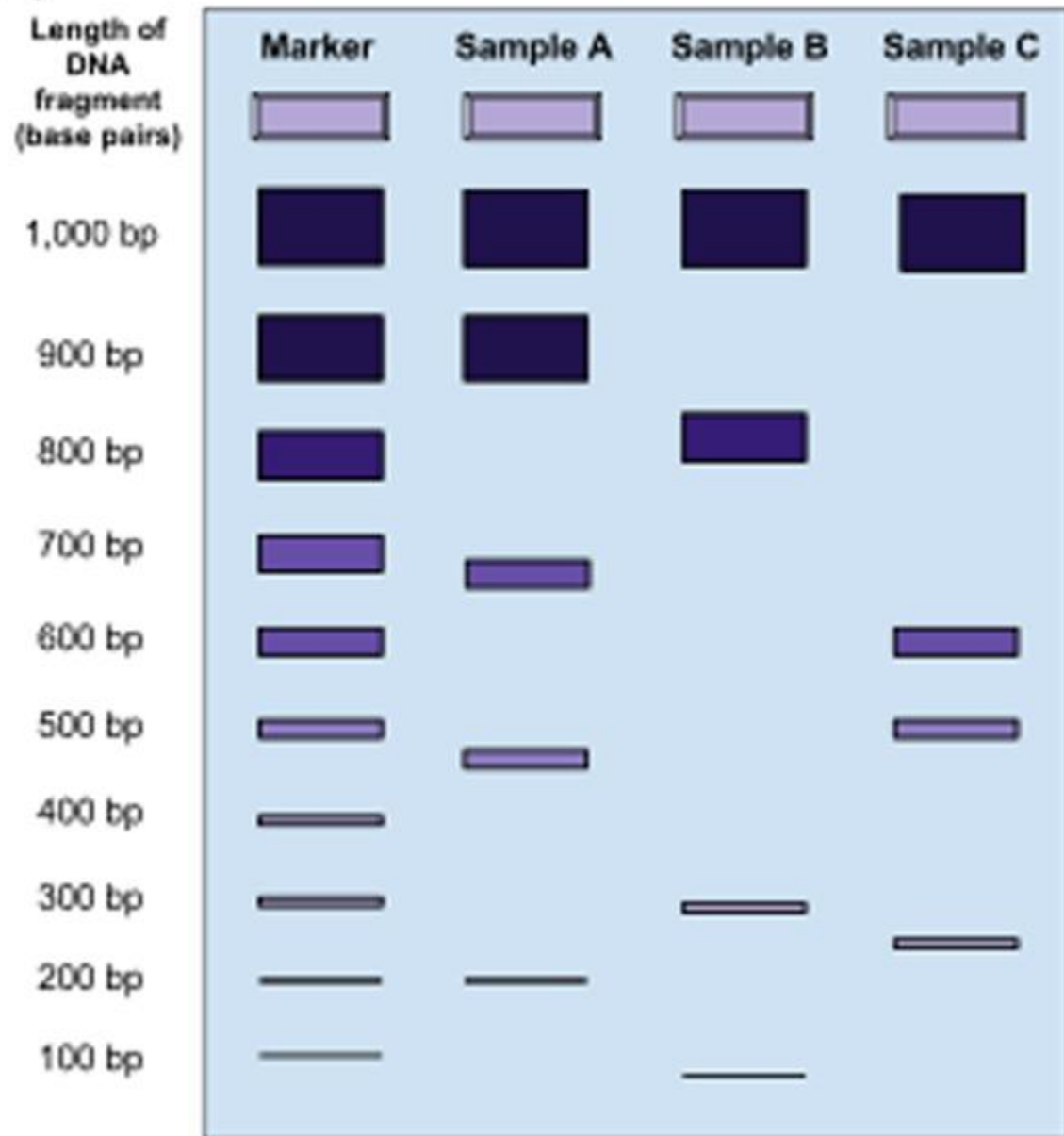
- This technique helps to separate DNA, RNA or proteins based on their size.
- For example, if you want to know the sizes of the DNA fragments after restriction enzyme digestion, you can use this procedure.
- Here, the DNA is run through a matrix or a gel and separated out. Agarose, a natural polymer from seaweeds is the most commonly used matrix. Let's understand how this technique works.



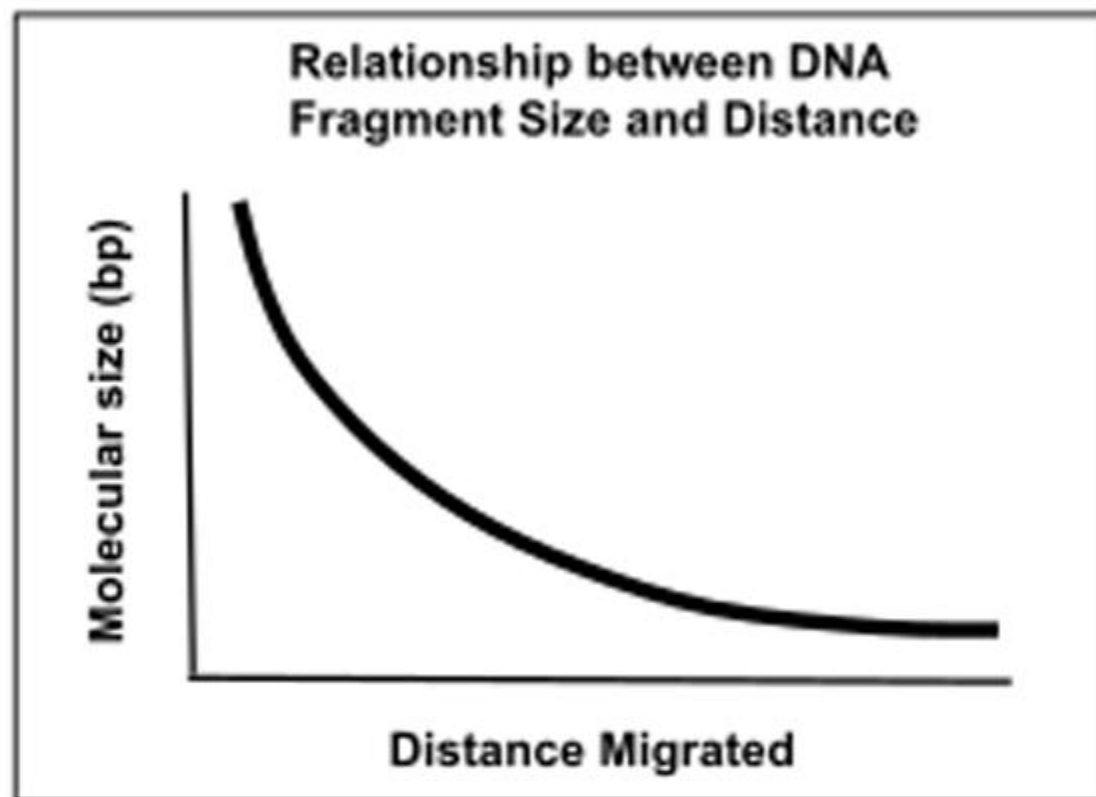


DNA electrophoresis

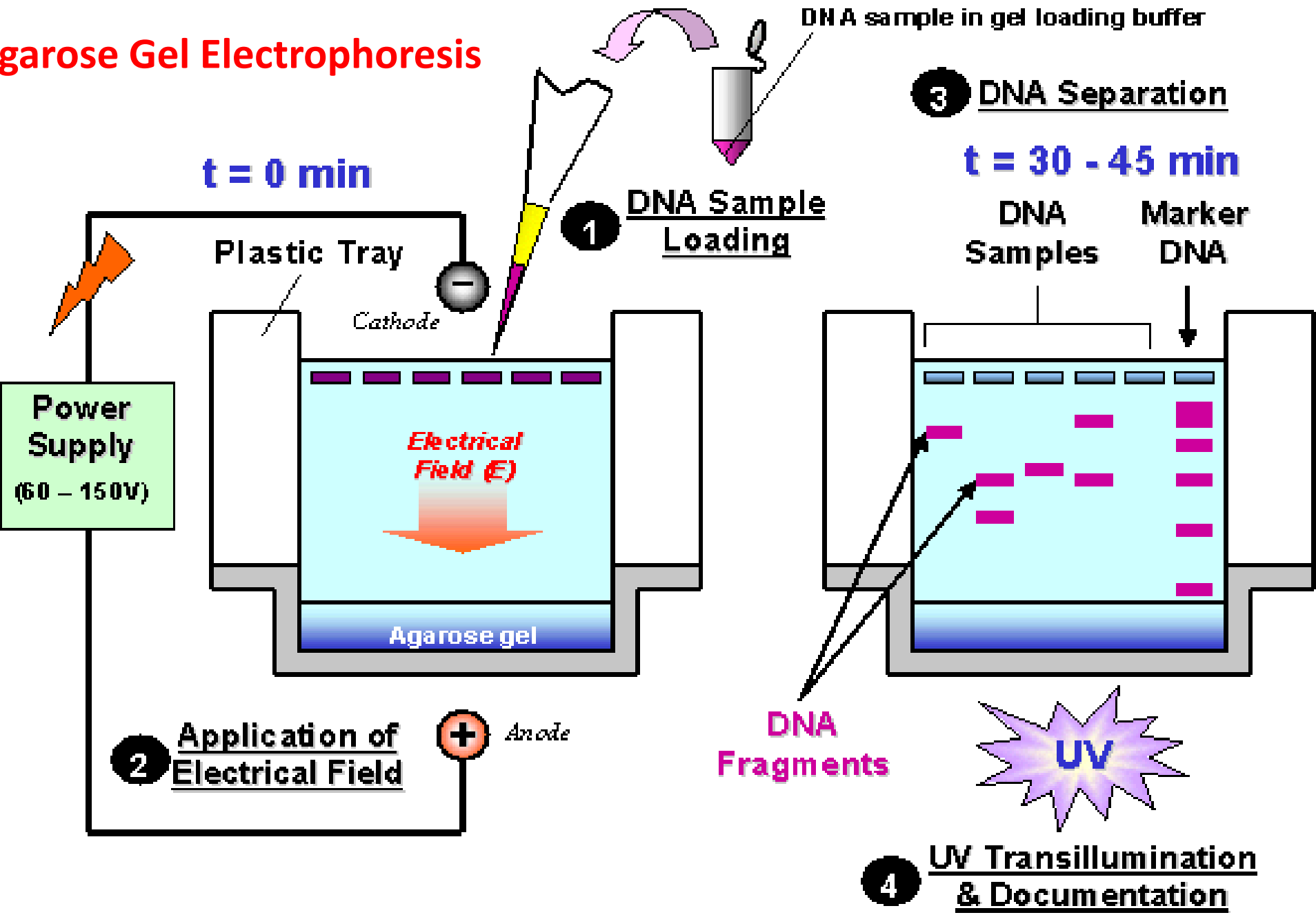
A)



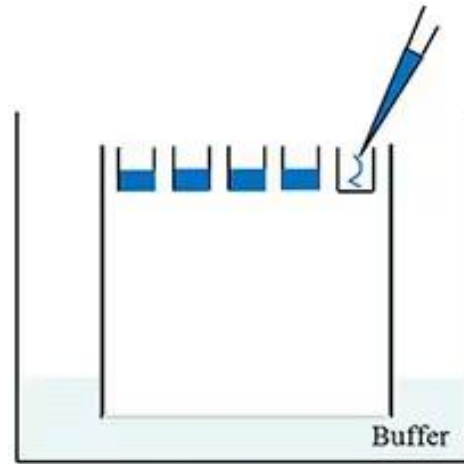
B)



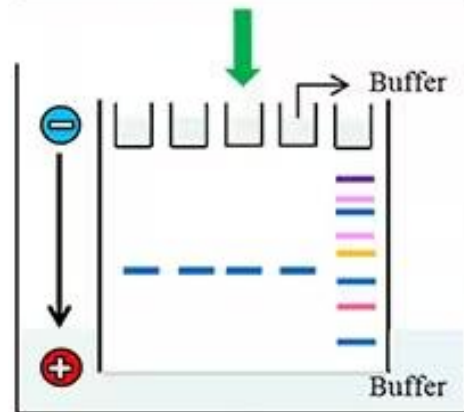
Steps Involved in Agarose Gel Electrophoresis



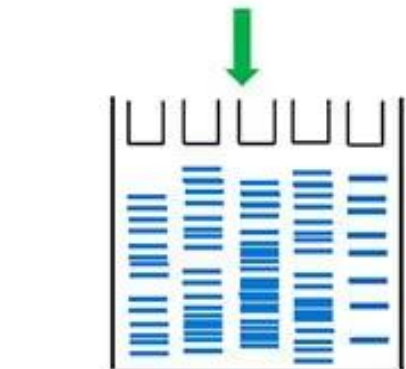
A schematic diagram of the complete process of SDS-PAGE



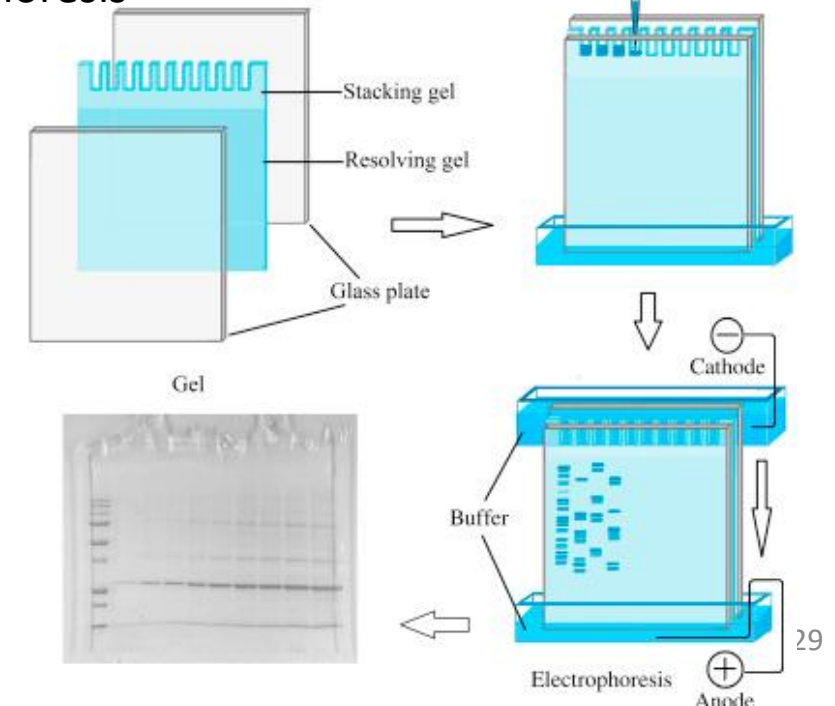
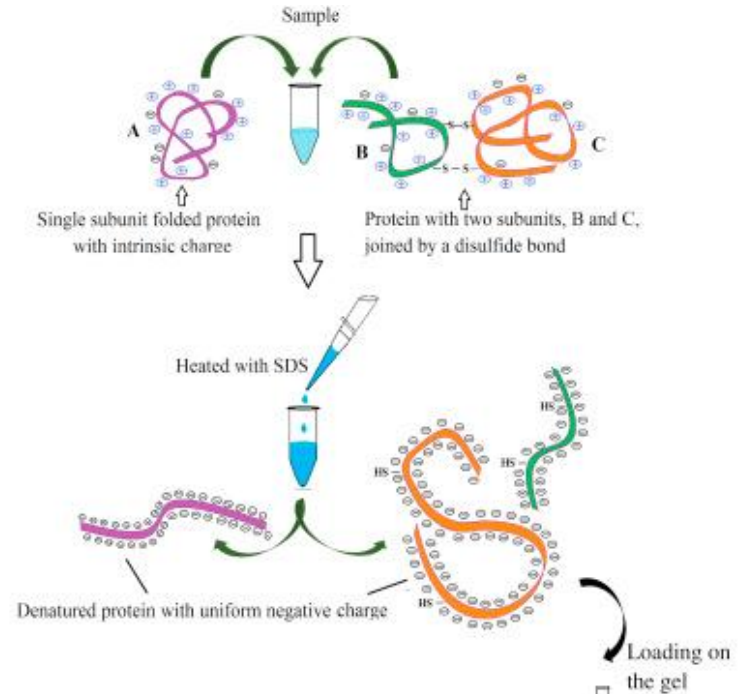
Protein samples and marker loaded in vertical SDS-PAGE system



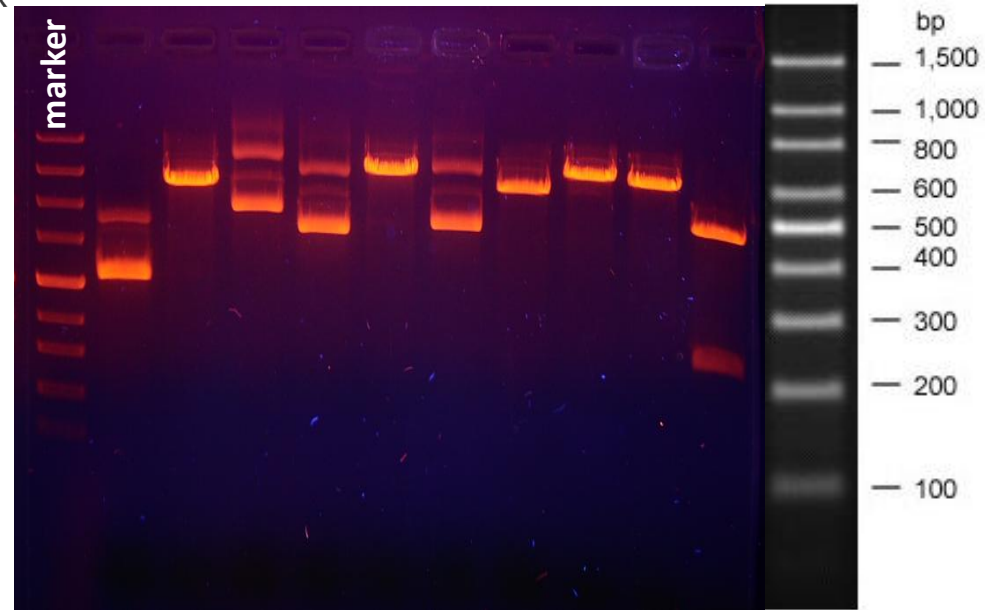
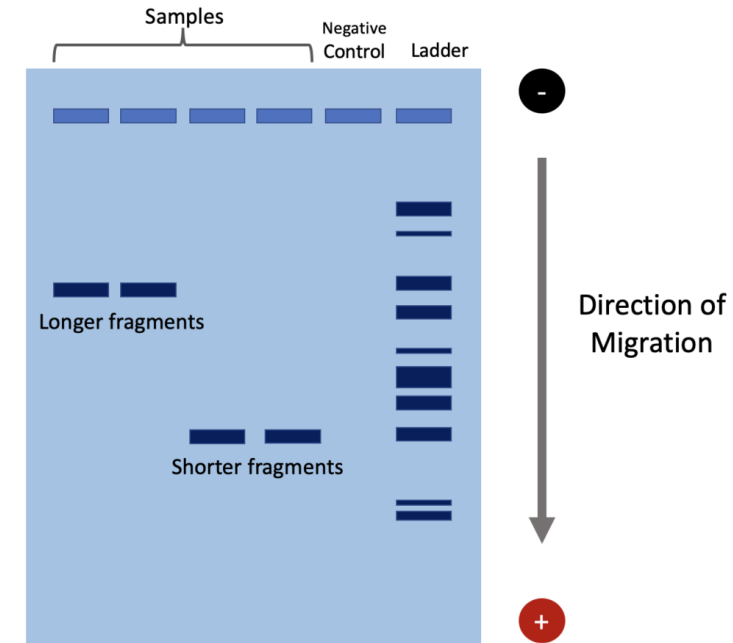
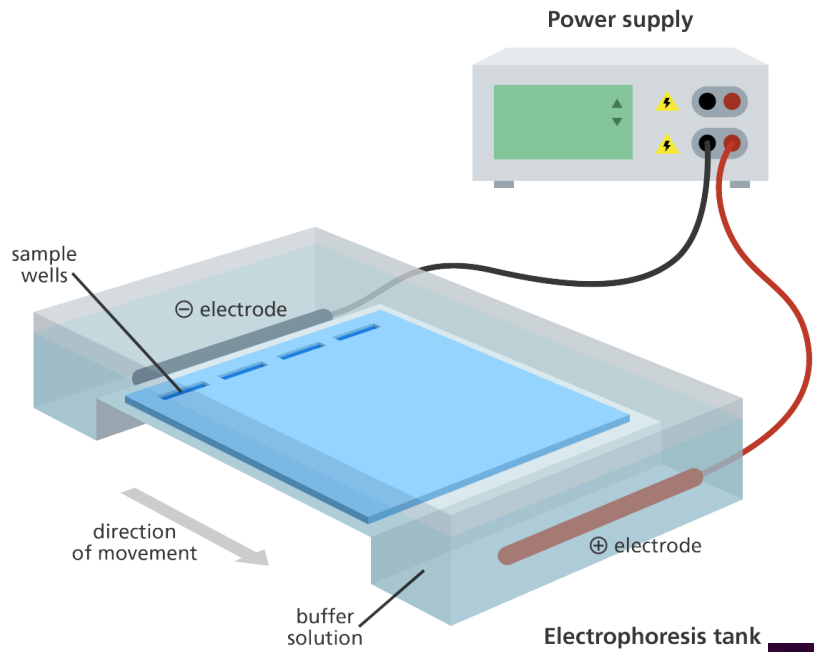
Steps Involved in Agarose Gel Electrophoresis



SDS-PAGE gel after Coomassie blue staining



Gel Electrophoresis



Gel Electrophoresis

- The DNA to be separated is added to the wells of an agarose gel matrix.
- On the application of current, the negatively charged DNA moves to the positive electrode.
- **Since the gel is difficult to move through, the DNA fragments travel at different speeds based on their size.**
- **Larger fragments move slower, while smaller fragments fit through the gel pores easily and move faster.**

Gel Electrophoresis

- The separated DNA fragments are not visible under normal light.
- visualized after staining the bands with a compound – ethidium bromide and exposing them to UV radiation.
- Once they are stained and exposed to UV light, you can see the separated DNA bands and determine their sizes using a DNA marker.
- In this way, we can cut out -
 - ✓ the desired size of DNA fragment
 - ✓ extract it and
 - ✓ further ligate it with cloning vectors.
- The techniques of PCR and gel electrophoresis are crucial in the field of forensics, for genetic fingerprinting and identifying crime suspects.

How?

DNA fingerprinting

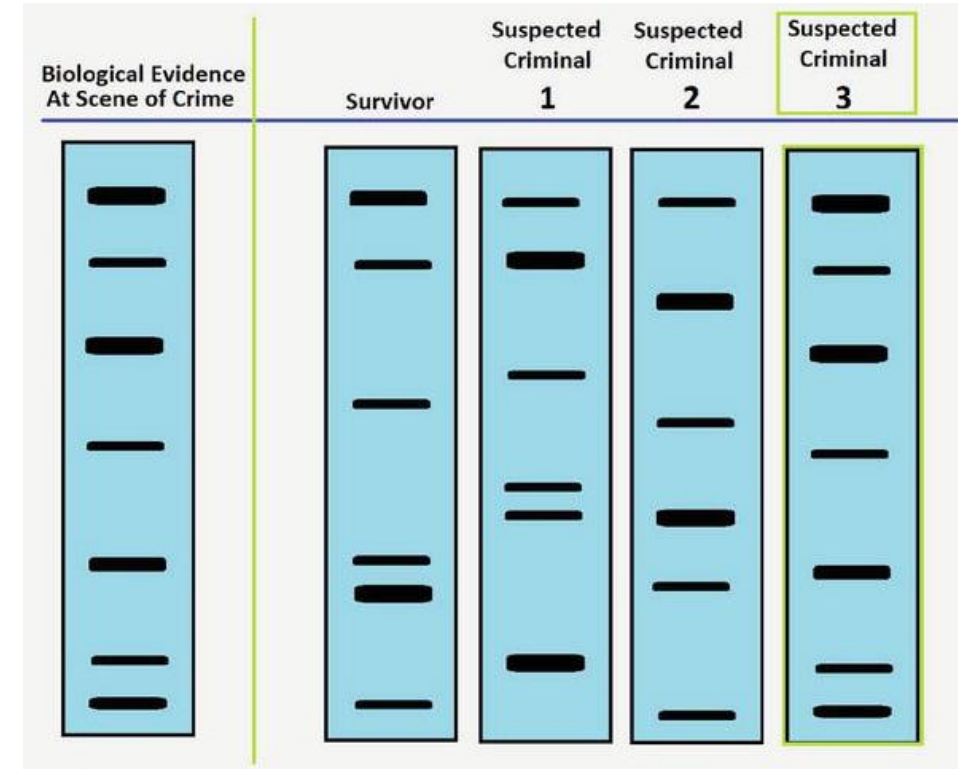
Restriction Fragment Length Polymorphism (RFLP) is a DNA fingerprinting technique that involves the following steps:

DNA extraction: DNA is extracted from a sample, such as blood or saliva, and purified.

DNA fragmentation: The purified DNA is cut into fragments using restriction endonucleases.

Gel electrophoresis: The DNA fragments are separated by size using gel electrophoresis.

Visualization: A labeled DNA sequence, called an RFLP probe, is used to reveal a unique pattern that identifies the DNA.



Recombinant DNA technology

Recombinant DNA technology involves the selection of the **desired gene** for administration into the host followed by a selection of the perfect vector with which the gene has to be integrated and hence the recombinant DNA is formed. This recombinant DNA, then has to be introduced into the **host cell (bacteria)**.

In standard cloning protocols, the **cloning of any DNA fragment** essentially involves **seven steps**:

- (1) Choice of host organism and cloning vector,
- (2) Preparation of vector DNA,
- (3) Preparation of DNA to be cloned,
- (4) Creation of recombinant DNA,
- (5) Introduction of recombinant DNA into the host organism,
- (6) Selection.

How does Biotechnology and Genetic Engineering works?

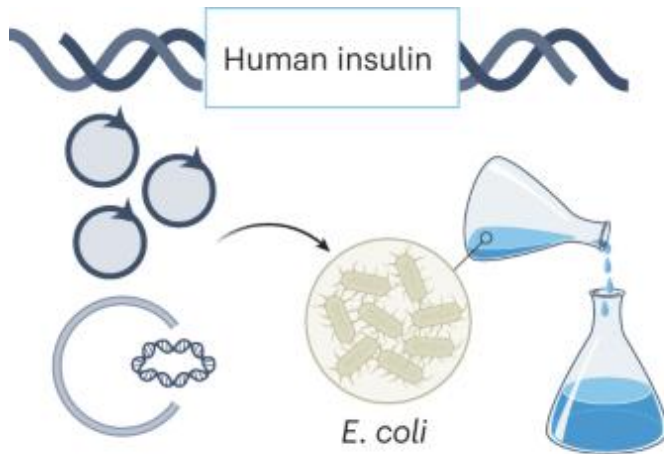
Genetic engineering is the cornerstone of modern biotechnology. It is based on scientific tools, developed in recent decades, that enable researchers to:

1. Identify the gene that produces the protein of interest.
2. Cut the DNA sequence that contains the gene from a sample of DNA.
3. Place the gene into a vector, such as a plasmid or bacteriophage.
4. Use the vector to carry the gene into the DNA of the host cells, such as *Escherichia coli* (*E coli*) or mammalian cells grown in culture.
5. Induce the cells to activate the gene and produce the desired protein.
6. Extract and purify the proteins
7. Quality control/assurance and marketing the products for therapeutic use.

Biotechnology: Principles and Processes

In biotechnology, scientists often need to amplify a small part of DNA or separate out different sized DNA fragments. How do scientists do this? They use specific techniques such as PCR and Gel electrophoresis that allow them to achieve these goals.

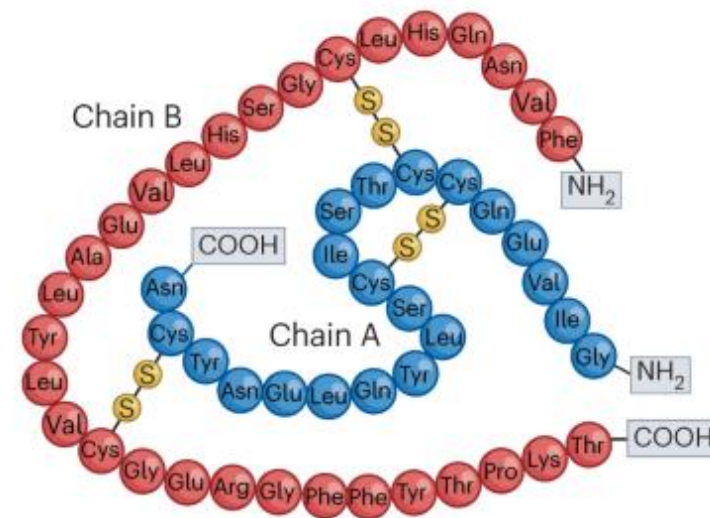
Let's learn about these techniques in detail.



Recombinant DNA technology



Large-scale fermentation

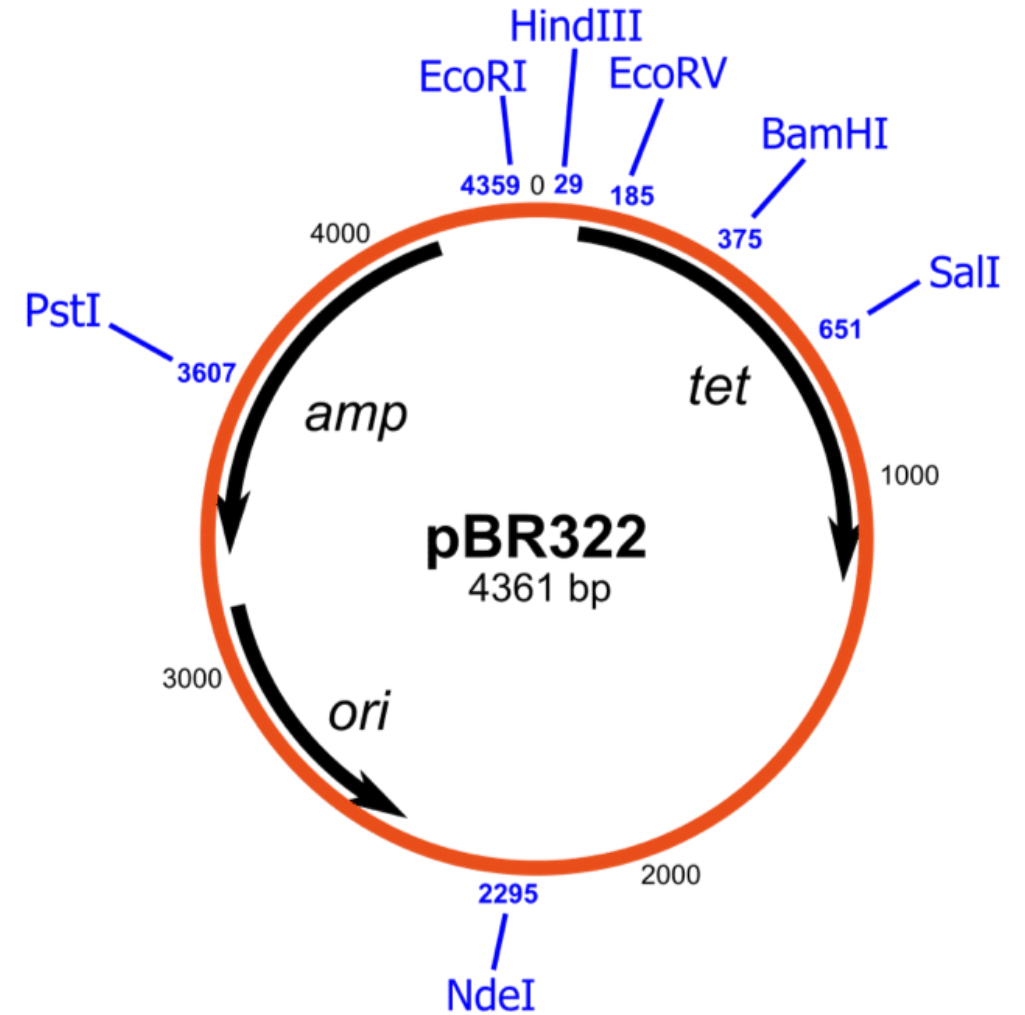
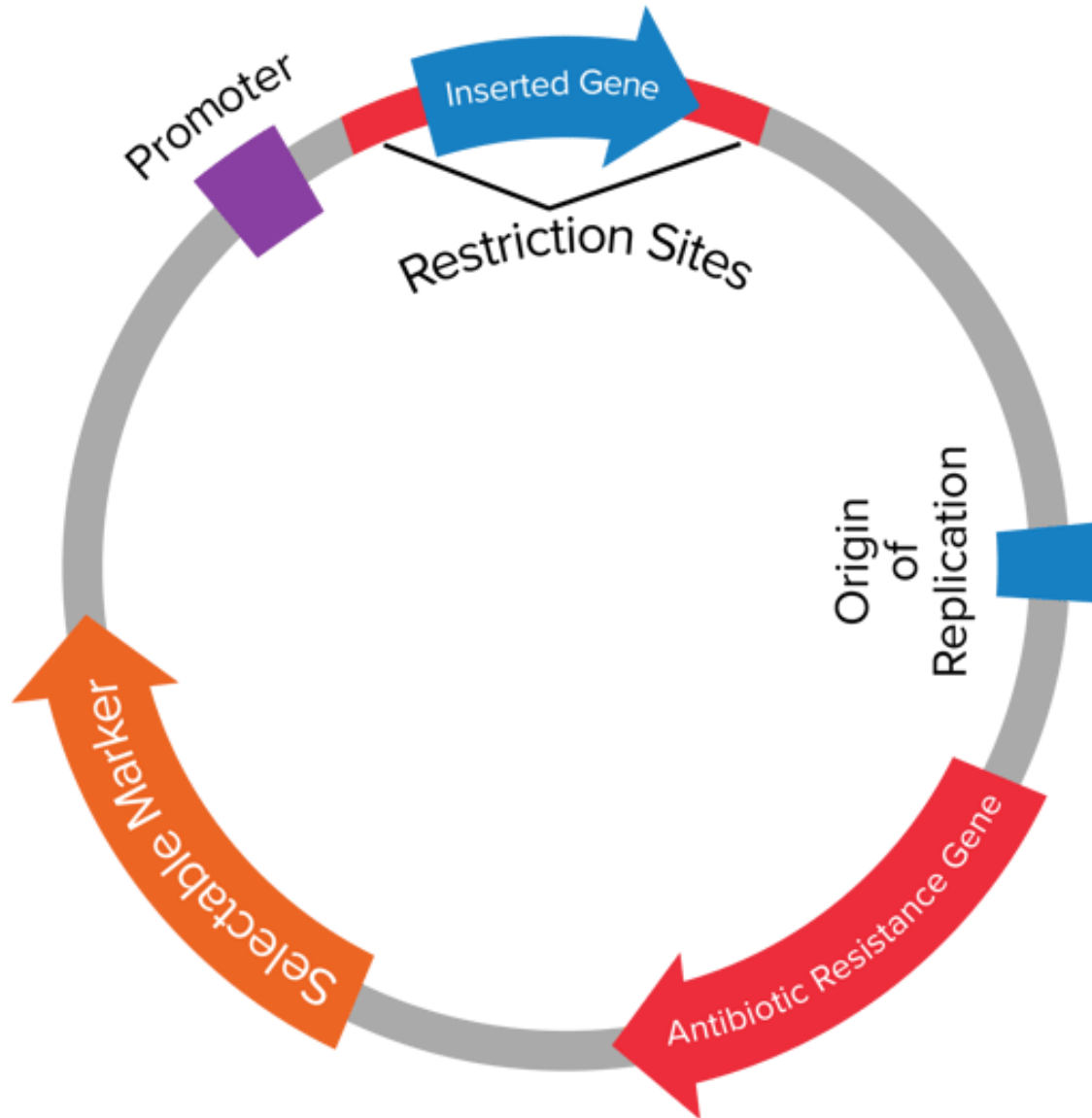


Insulin harvested and purified

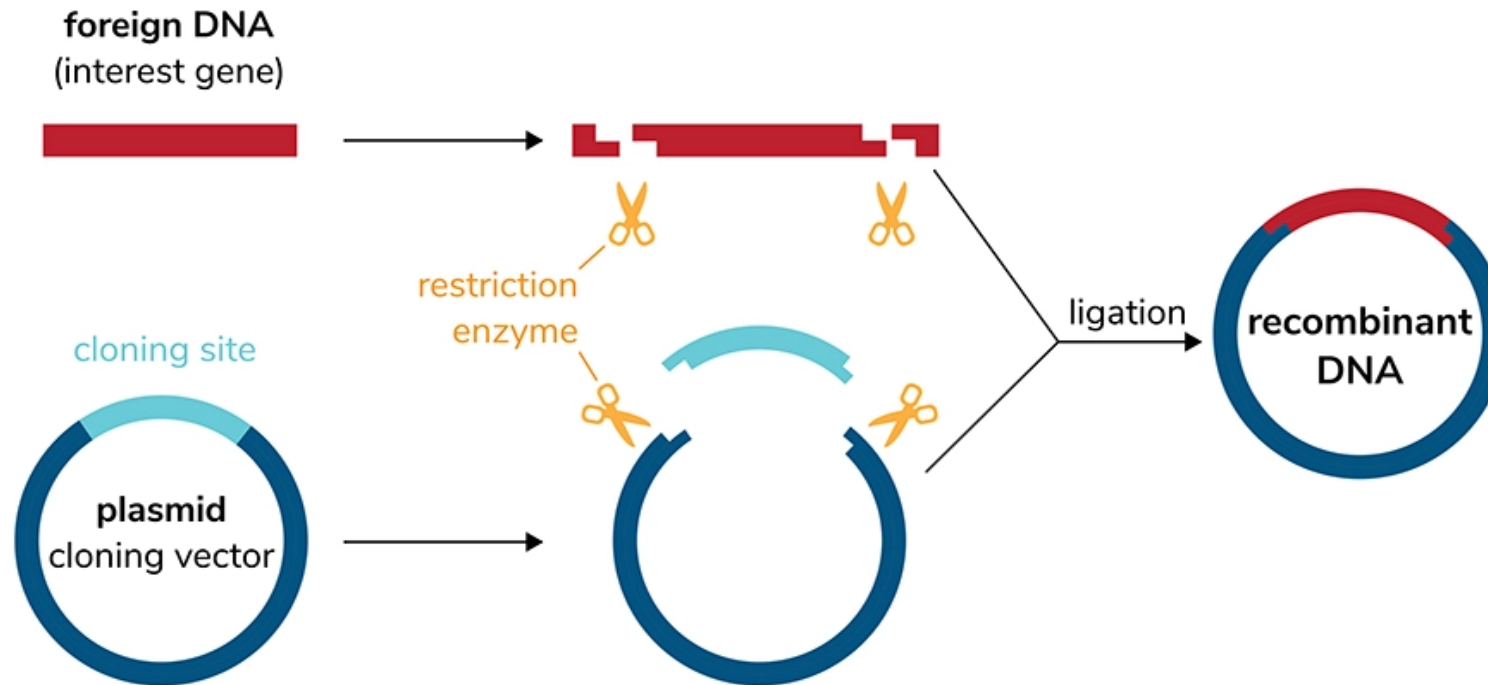


Commercial insulin

A Plasmid/Vector map (Career/ Shuttle to transport your DNA)



Recombinant DNA (rDNA)

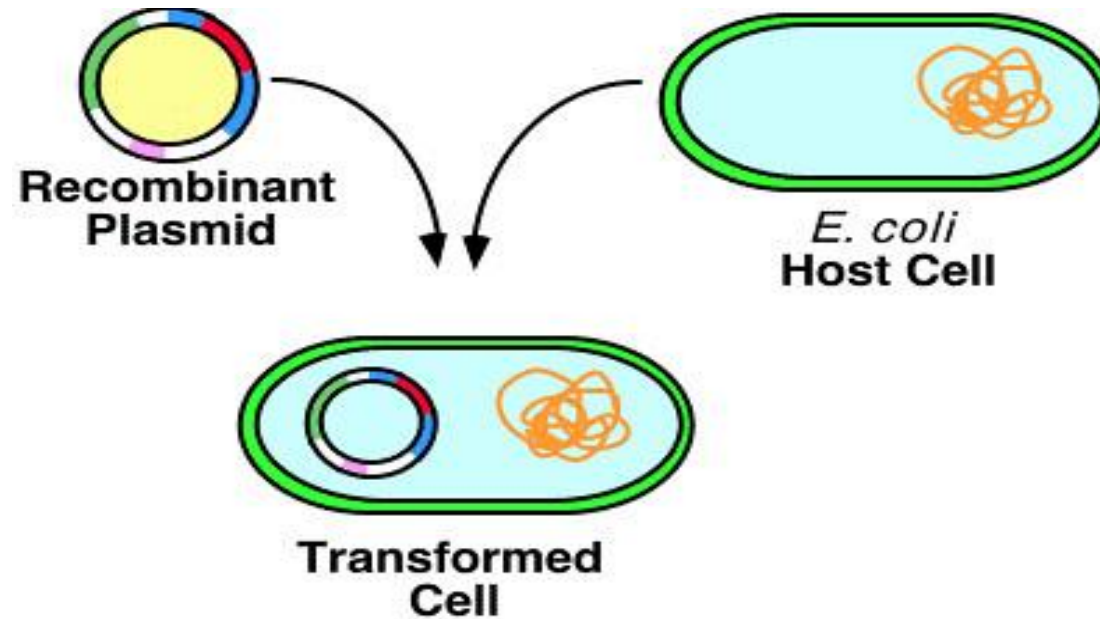


Recombinant DNA (rDNA) is a piece of DNA that's created by combining two or more DNA fragments from different sources. It's possible to create rDNA because DNA molecules from all organisms share the same chemical structure, but differ in the nucleotide sequence

The process of creating rDNA involves:

1. Inserting a DNA fragment into a small DNA molecule, called a DNA vector
2. Allowing the DNA vector to replicate inside a living cell, such as a bacterium
3. Generating multiple copies of the DNA fragment, creating a DNA clone

Host cell/Competent cell/ 3rd party producer, manufacturer

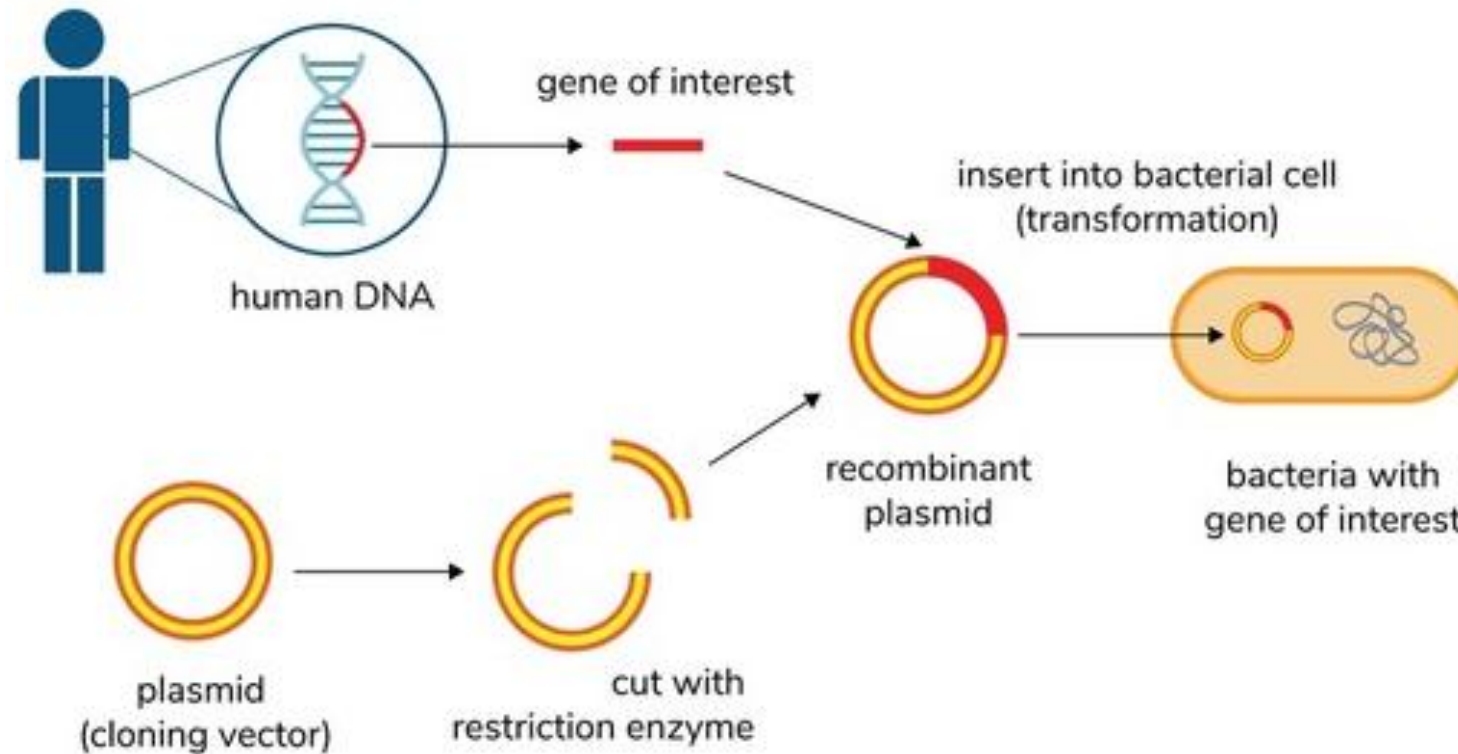


A host cell in cloning is a living cell that can propagate a vector, or carrier molecule, that contains a desired gene. The host cell must be able to:

- Allow the recombinant DNA (rDNA) to easily enter the cell
- Not degrade the rDNA as foreign DNA
- Provide the enzymes and proteins needed for the vector DNA to replicate

In cloning applications, this new DNA molecule is introduced into a host cell for replication and expression of the desired gene product. Bacterial host cells, such as *Escherichia coli* (*E. coli*), are commonly used for this purpose because they grow rapidly, are easy to manipulate, and have well-understood genetics

Recombinant **DNA** technology



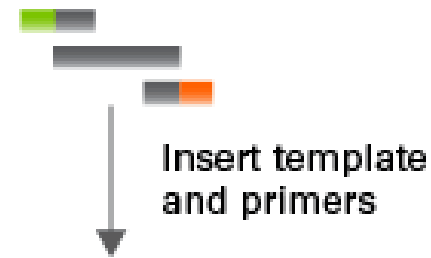
Recombinant DNA technology involves using enzymes and various laboratory techniques to manipulate and isolate DNA segments of interest. This method can be used to combine (or splice) DNA from different species or to create genes with new functions. The resulting copies are often referred to as recombinant DNA. Such work typically involves propagating the recombinant DNA in a bacterial or yeast cell, whose cellular machinery copies the engineered DNA along with its own.

3 Simple steps in Recombinant **DNA** technology

1

Amplify

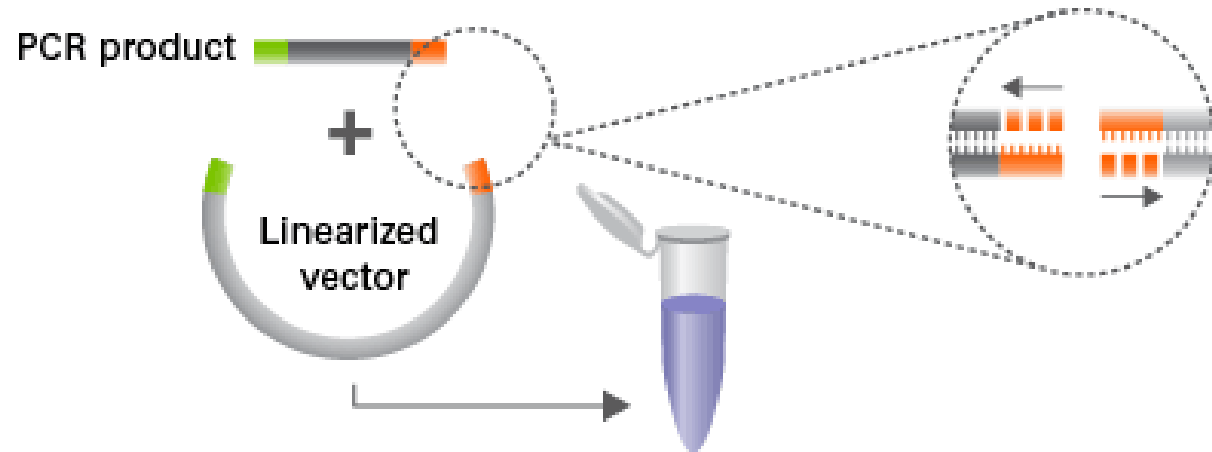
each fragment as a PCR product with 15-bp extensions complementary to the vector ends.



2

Combine

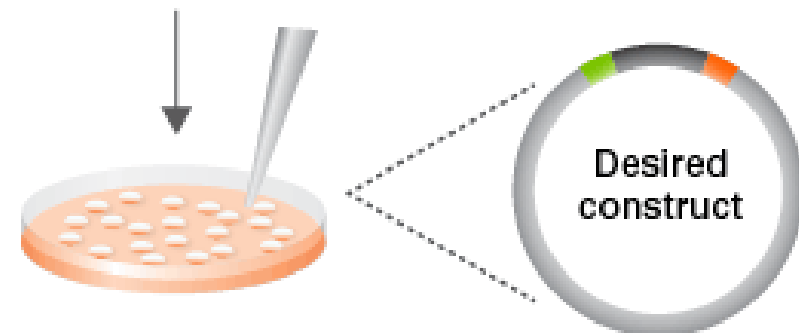
the In-Fusion master mix, linearized vector, and PCR product in a single-tube, 15-min cloning reaction.



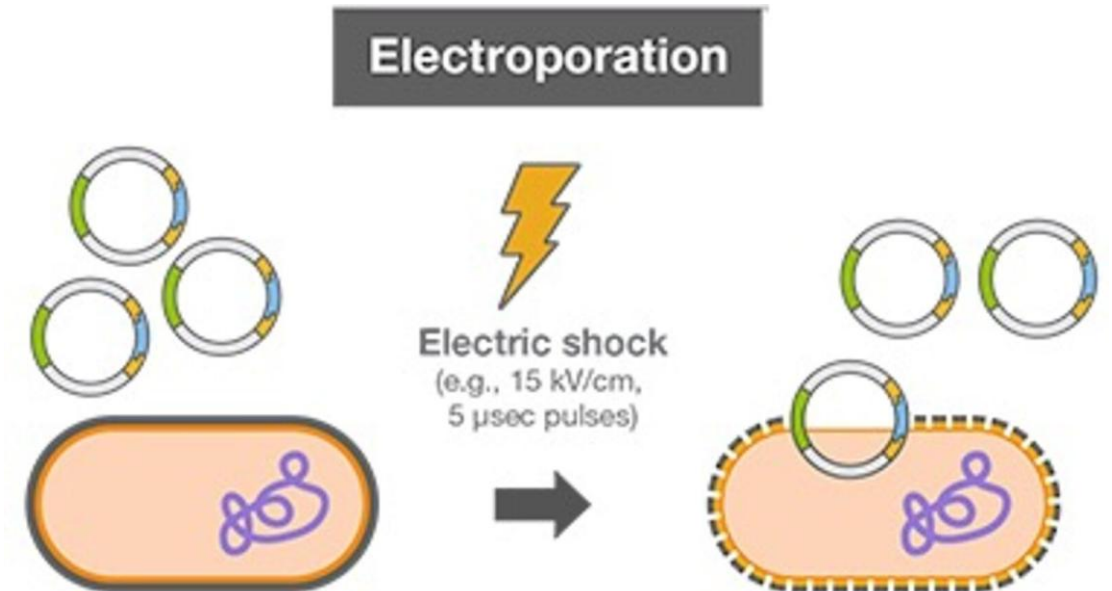
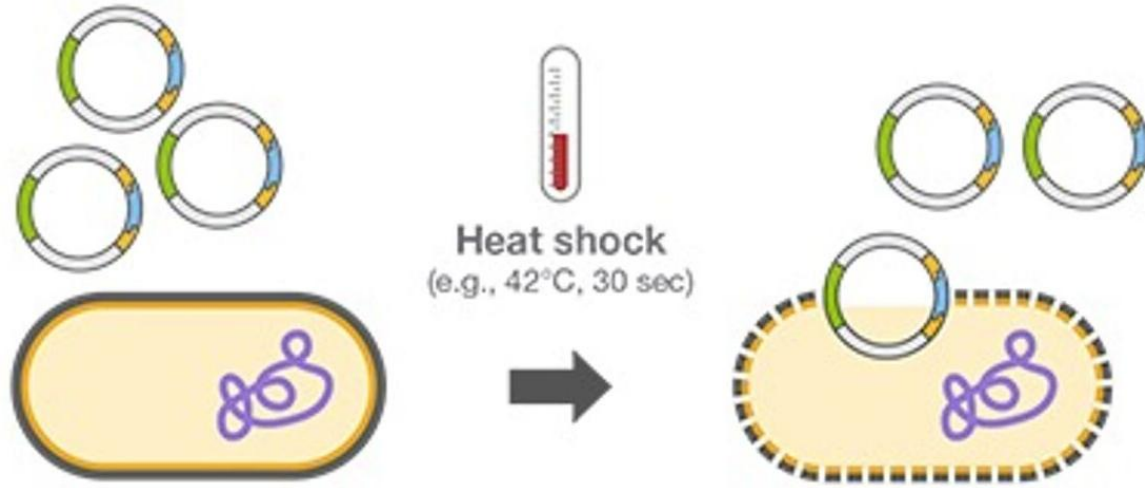
3

Transform

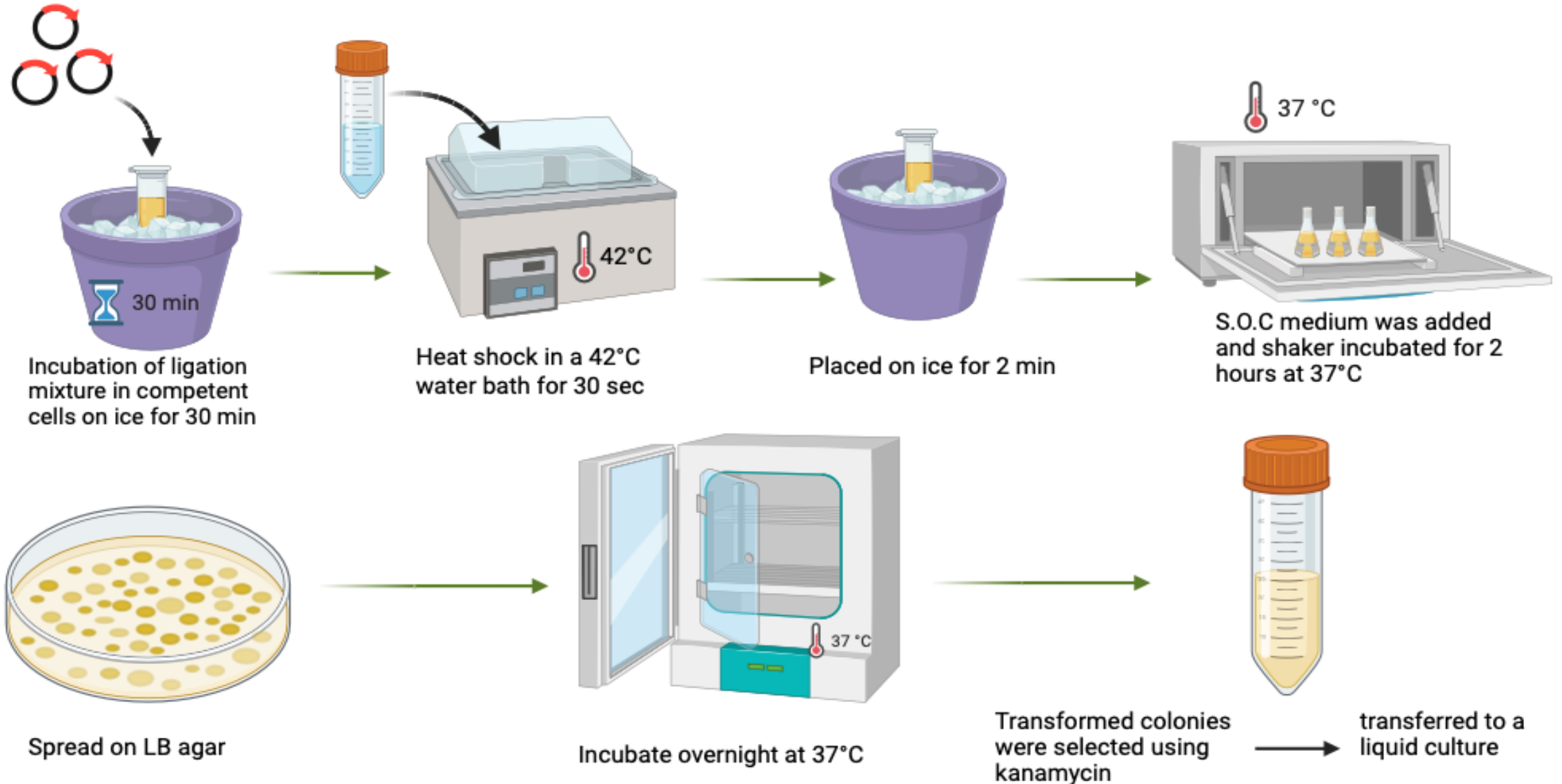
competent cells with the cloning reaction. Plate on selective media.



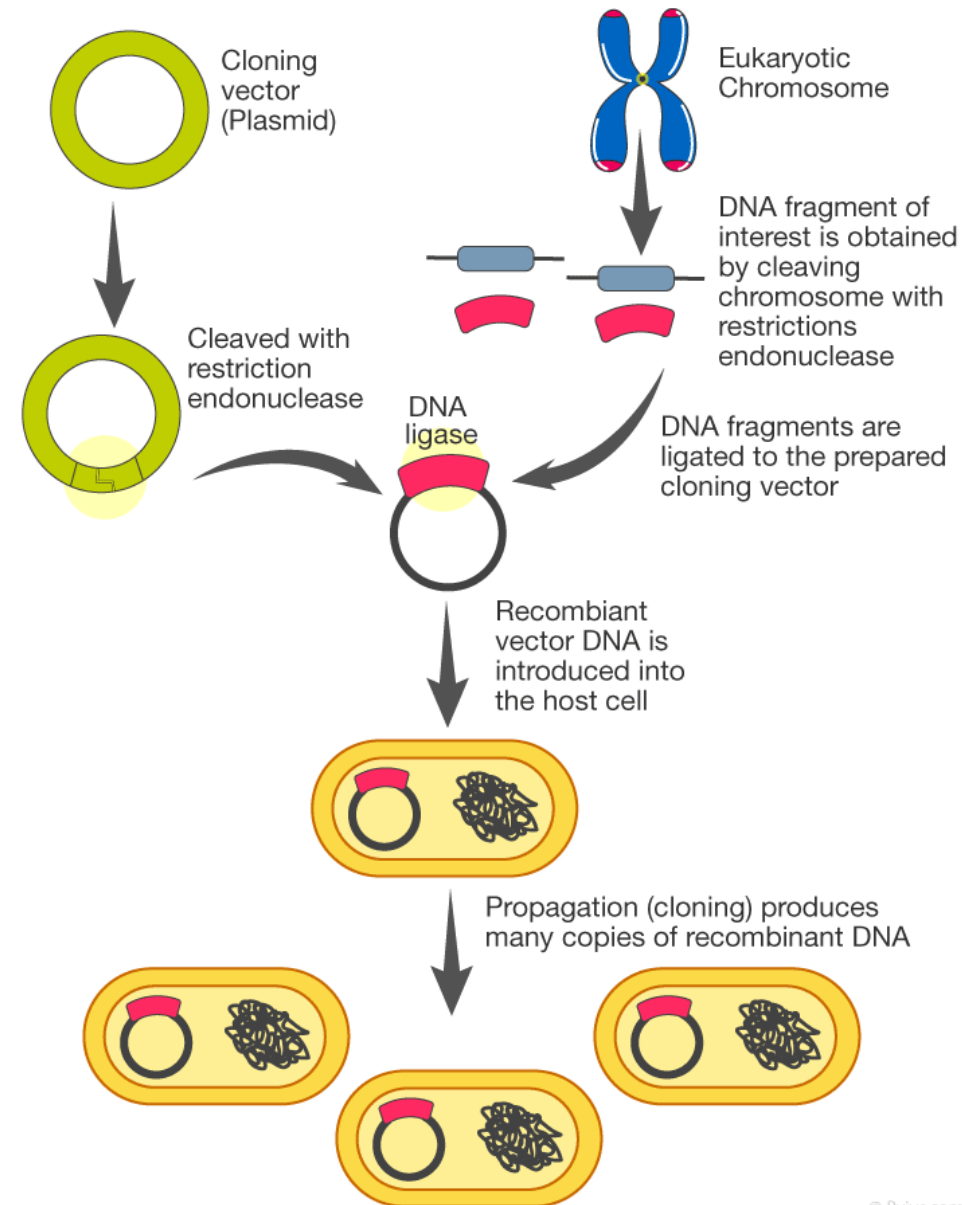
Insertion of foreign gene into Bacteria



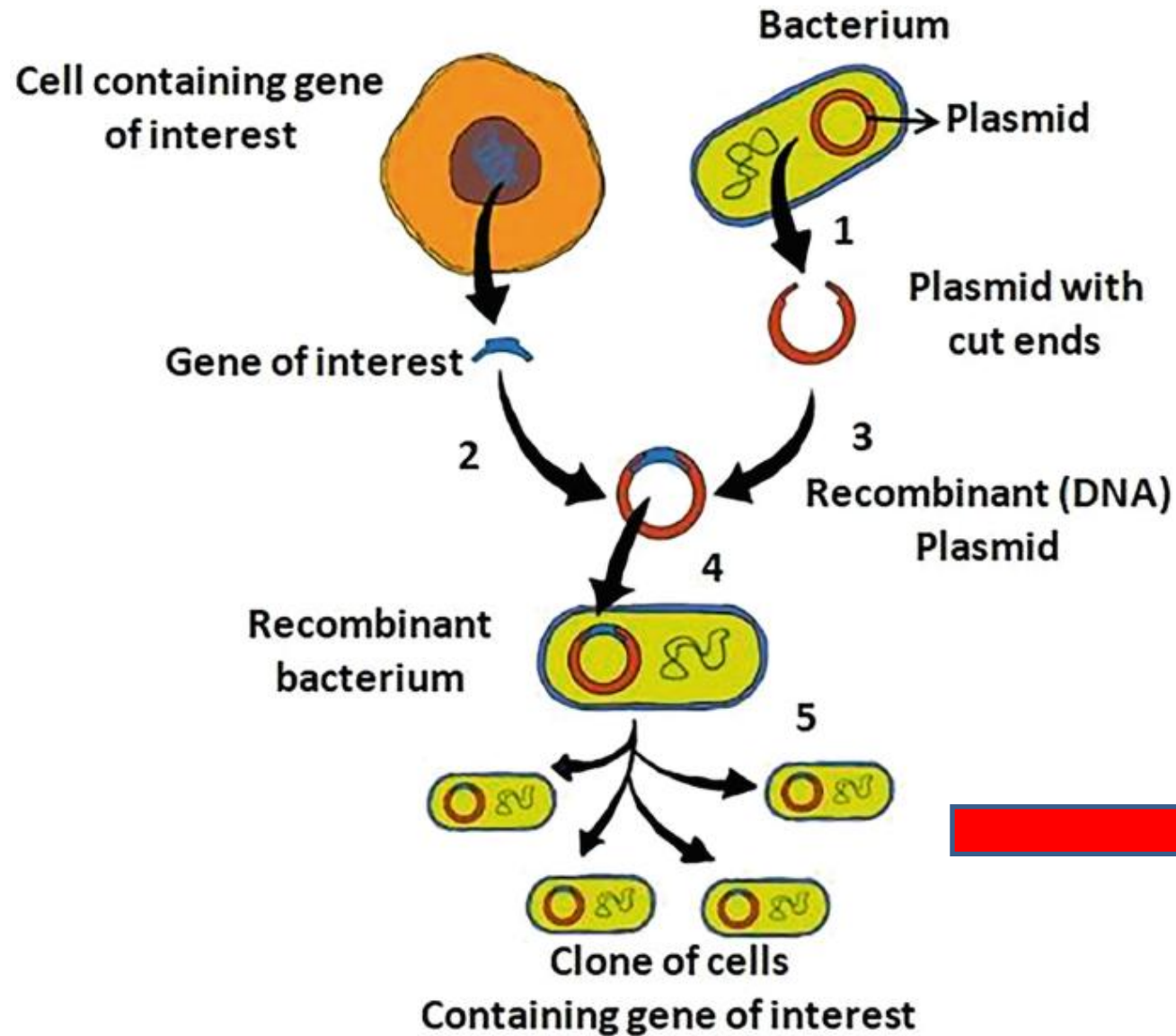
Overview of bacterial Transformation (insertion of foreign gene into Bacteria/ E.coli)



Recombinant DNA technology



Expansion of your bacteria carrying foreign genes

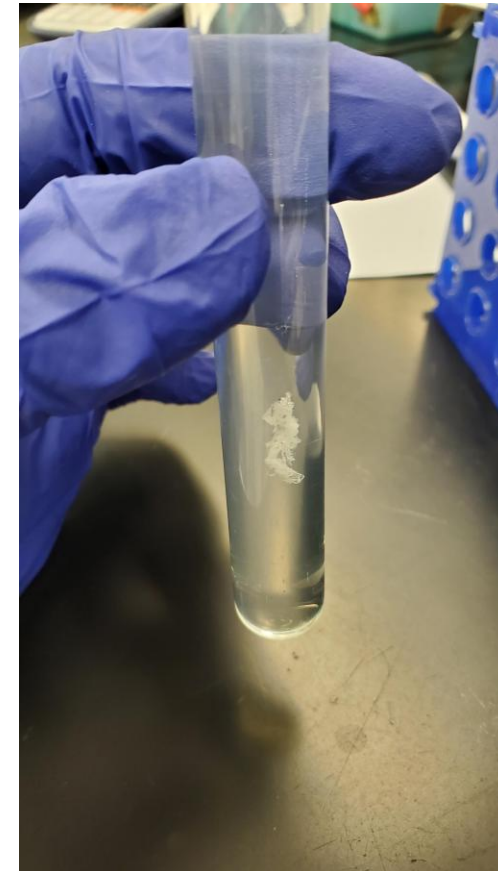
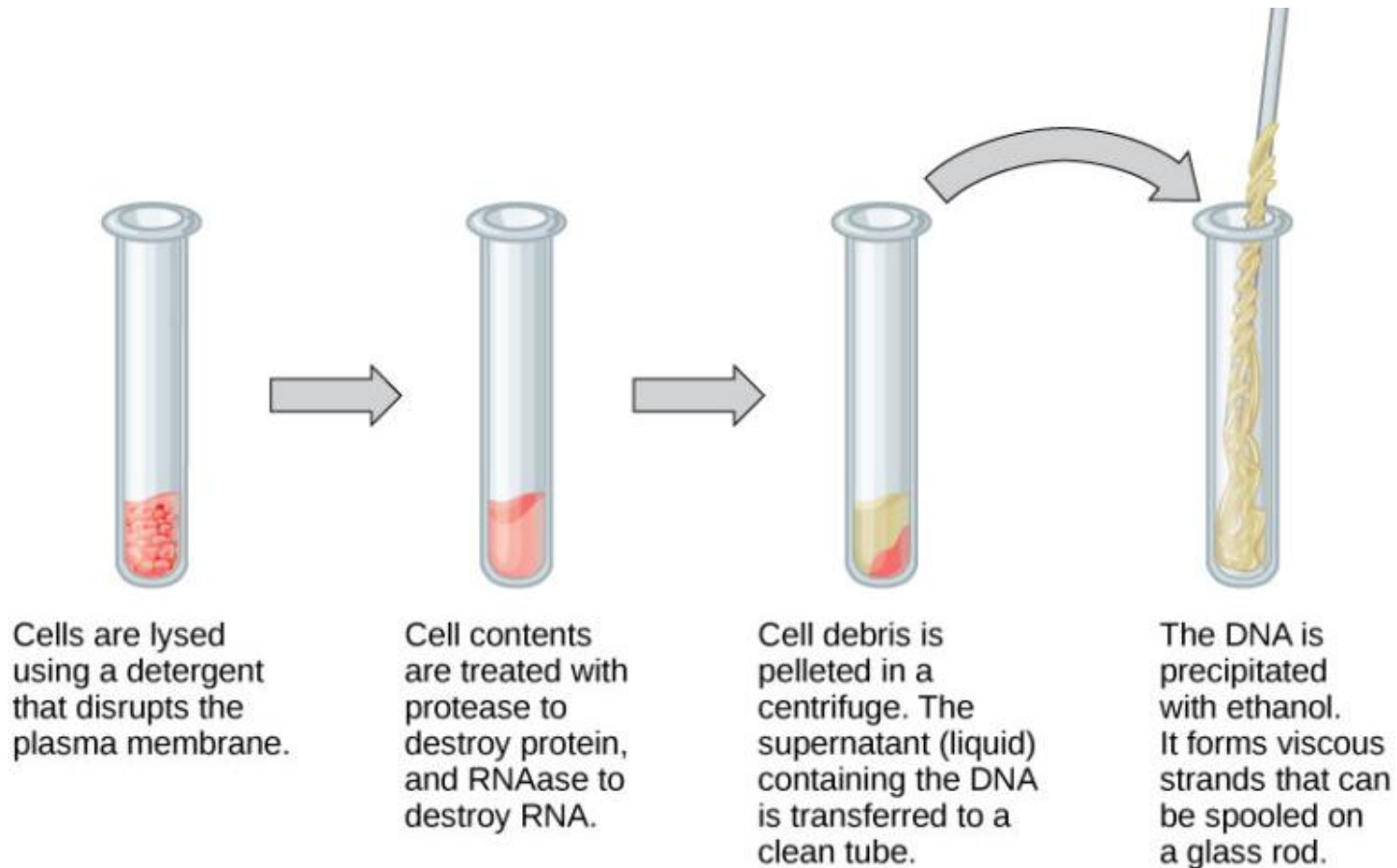


Workflow of rDNA technology

The **first step** in rDNA technology

1. Isolation of the desired DNA in its pure form i.e. free from other macromolecules.

DNA extraction:



Workflow of rDNA technology

However, in a normal cell, the DNA not only exists within the cell membrane, but also present along with other macromolecules such as RNA, polysaccharides, proteins, and lipids.

So, how do we break open the cell and obtain DNA that is free from other macromolecules?

Some enzymes are here to help

Lysozyme – to break bacterial cell wall.

Cellulase – to break plant cell wall.

Chitinase – to break fungal cell wall.

Ribonuclease – removes RNA.

Protease – removes proteins (such as histones that are associated with DNA).

Workflow of rDNA technology

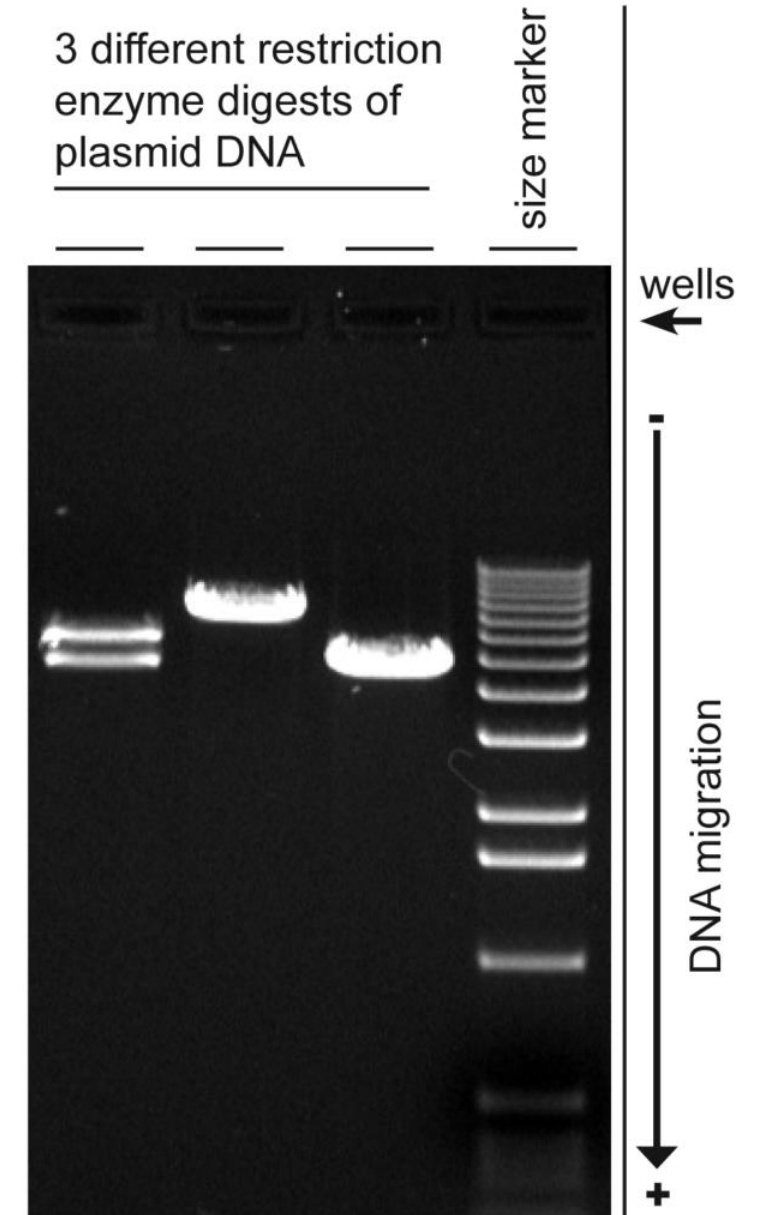
2. Restriction Enzyme Digestion

Restriction enzymes act as molecular scissors : cut DNA at specific locations.

These reactions are called 'restriction enzyme digestions.

They involve the incubation of the purified DNA with the selected restriction enzyme, at conditions optimal for that specific enzyme.

The technique – 'Agarose Gel Electrophoresis' reveals the progress of the restriction enzyme digestion. This allows us to separate and cut out the digested DNA fragments. The vector DNA is also processed using the same procedure.



Tools of Biotechnology: Restriction Enzymes

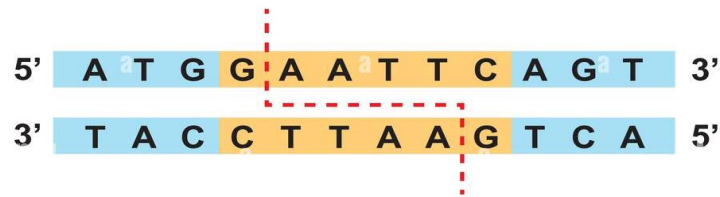
- Restriction enzymes, also known as 'restriction endonucleases' are **molecular scissors** that can cut DNA at specific locations. These are part of a larger class of enzymes called 'Nucleases'. Nucleases are of two kinds:
 - Exonucleases** – Enzymes that remove nucleotides from the **ends of DNA**.
 - Endonucleases** – Enzymes that make cuts at **specific positions in the DNA**.
- '*Hind II*' – the first restriction endonuclease to be isolated, identifies a specific six base pair sequence and always cuts the DNA at a particular point. This specific sequence is the '**Recognition sequence**'.
- Today, there are more than 900 restriction enzymes, isolated from about 230 bacterial strains. Each of these enzymes recognizes different recognition sequences.

Tools of Biotechnology: Restriction Enzymes

Mode of Action of Restriction Enzymes

- Inspect the length of DNA for the specific recognition sequence.
- Bind the DNA at the recognition sequence.
- Cut the two strands of DNA at specific points in their sugar-phosphate backbone.
- Restriction enzymes recognize specific palindromic nucleotide sequences in the DNA.

EcoRI Enzyme Restriction Site



Palindromic Sequence

MADAM IM ADAM

A TOYOTA

Workflow of rDNA technology

3. Amplification Using PCR

PCR is a method of making multiple copies of a DNA sequence using the enzyme – DNA polymerase. It helps to amplify a single copy or a few copies of DNA into thousands to millions of copies. PCR reactions are run on 'thermal cyclers' using the following components:

Template – DNA to be amplified

Primers – small, chemically synthesized oligonucleotides that are complementary to a region of the DNA.

Enzyme – DNA polymerase

Nucleotides – needed to extend the primers by the enzyme.

The cut fragments of DNA can be amplified using PCR and then ligated with the cut vector as explained in upcoming slides.

Workflow of rDNA technology

4. Ligation of DNA Molecules

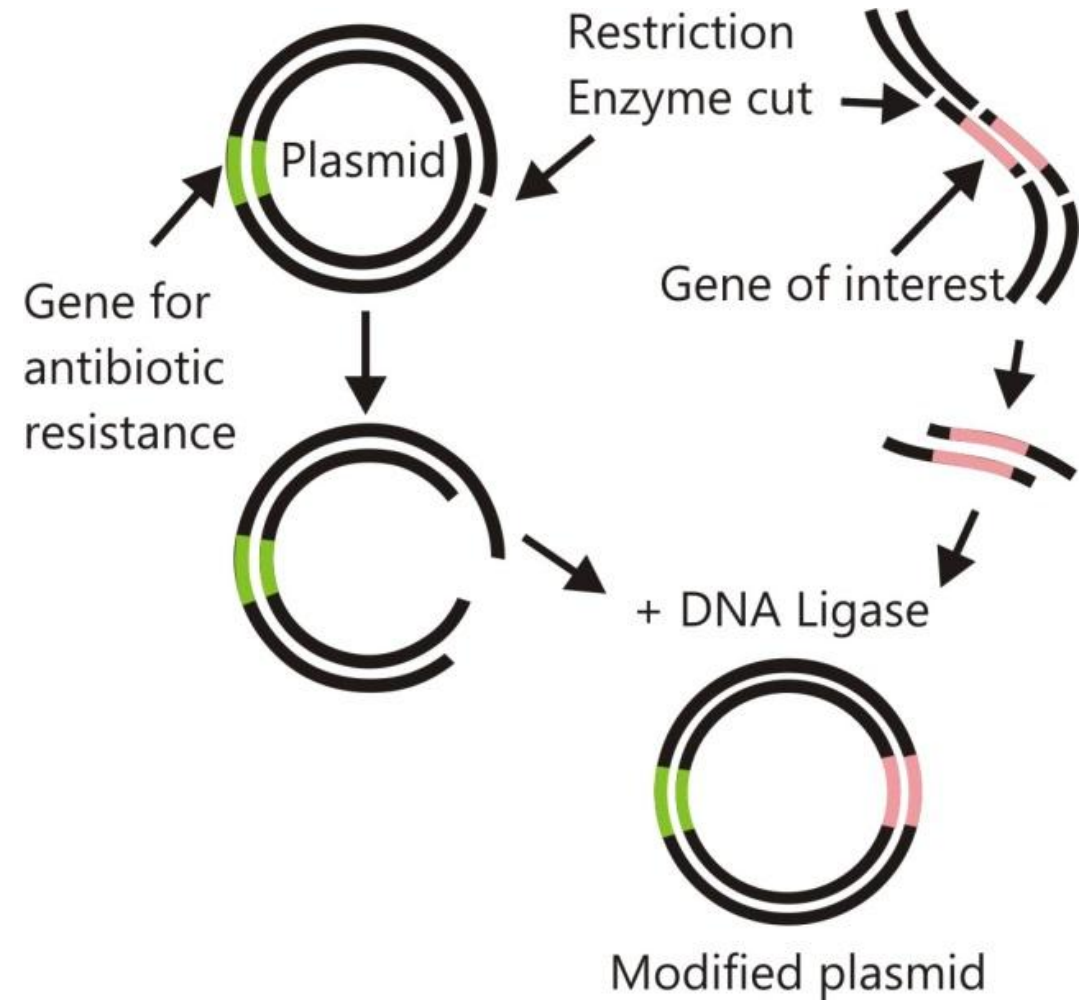
The purified DNA and the vector of interest :cut with the same restriction enzyme.

This gives us the cut fragment of DNA and the cut vector, that is now open.

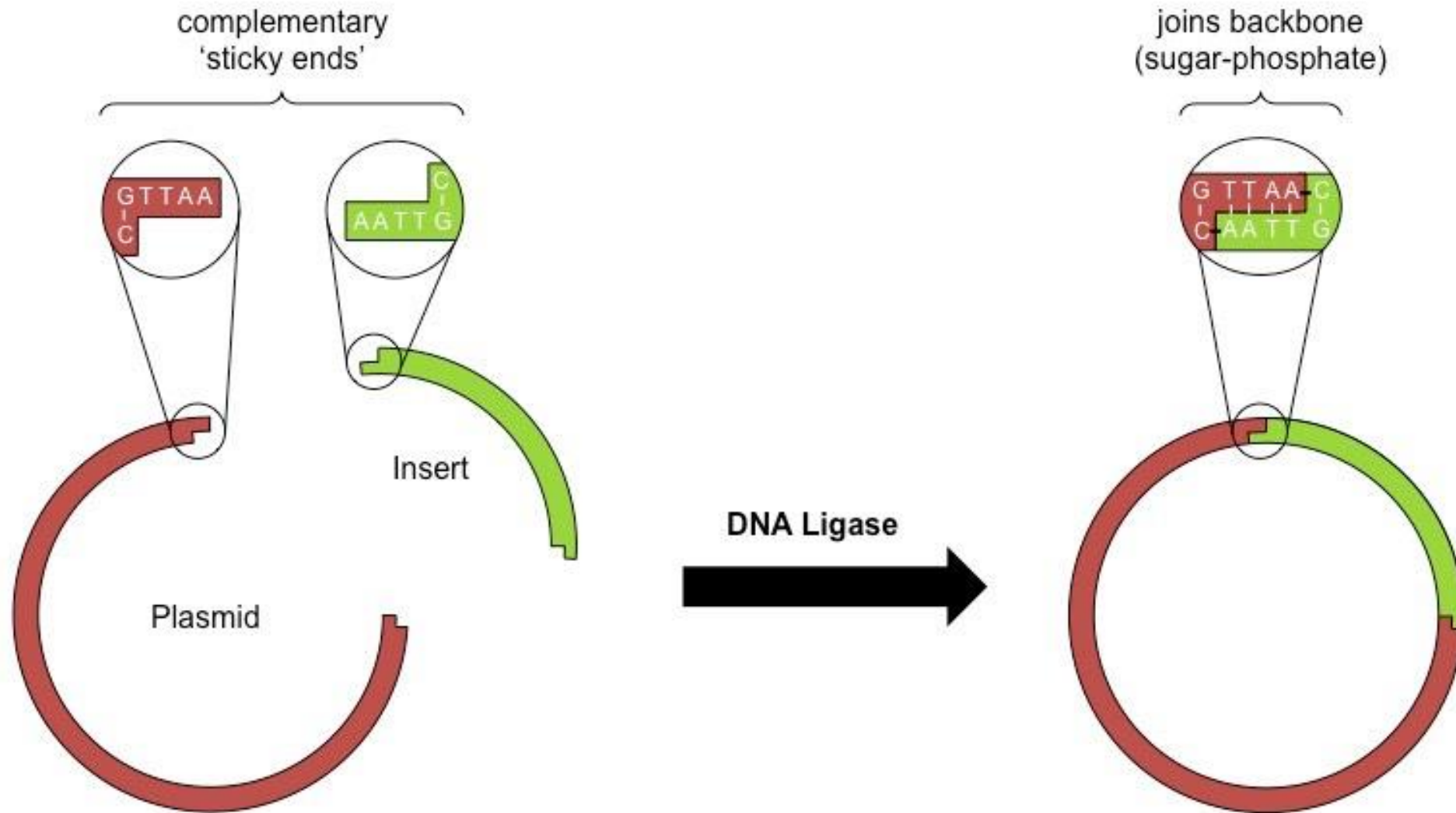
The process of joining these two pieces together using the enzyme 'DNA ligase' is 'ligation'.

The resulting DNA is '**recombinant DNA**'.

In the diagram, a desired gene for antibiotic resistance is ligated to vector for insertion.



Ligation of Vector and Insert



Workflow of rDNA technology

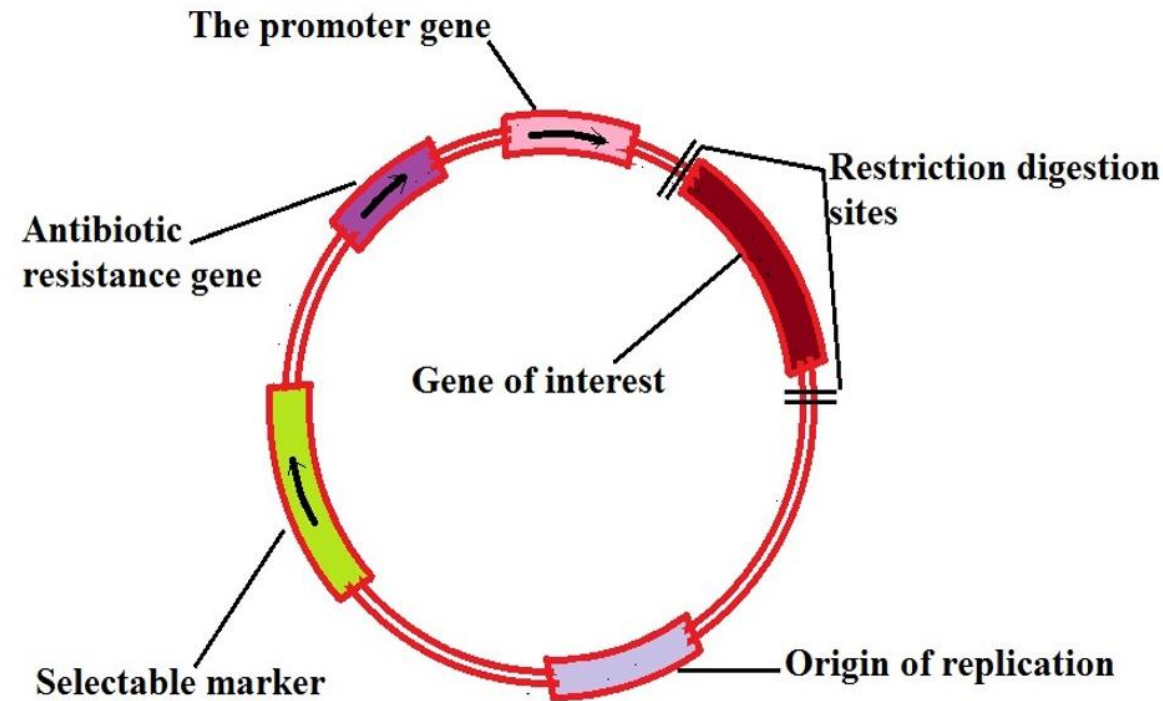
5. Insertion of Recombinant DNA Into Host Cells (bacteria):

In this step, the recombinant DNA is introduced into a recipient host cell. This process is called ‘Transformation’. Bacterial cells do not accept foreign DNA easily. Therefore, they are treated to make them ‘**competent**’ to accept new DNA.

During transformation, if a recombinant DNA bearing a gene for ampicillin resistance is transferred into recipient E. coli cells, then the E. coli cells also become ampicillin-resistant. This aspect is useful in differentiating transformed cells from non-transformed cells.

For example, if we spread the transformed cells on agar plates containing ampicillin, only the transformed, ampicillin-resistant cells will grow while the untransformed cells will die. Therefore, in this case, the ampicillin resistance gene acts as the ‘selectable marker’.

Workflow of rDNA technology

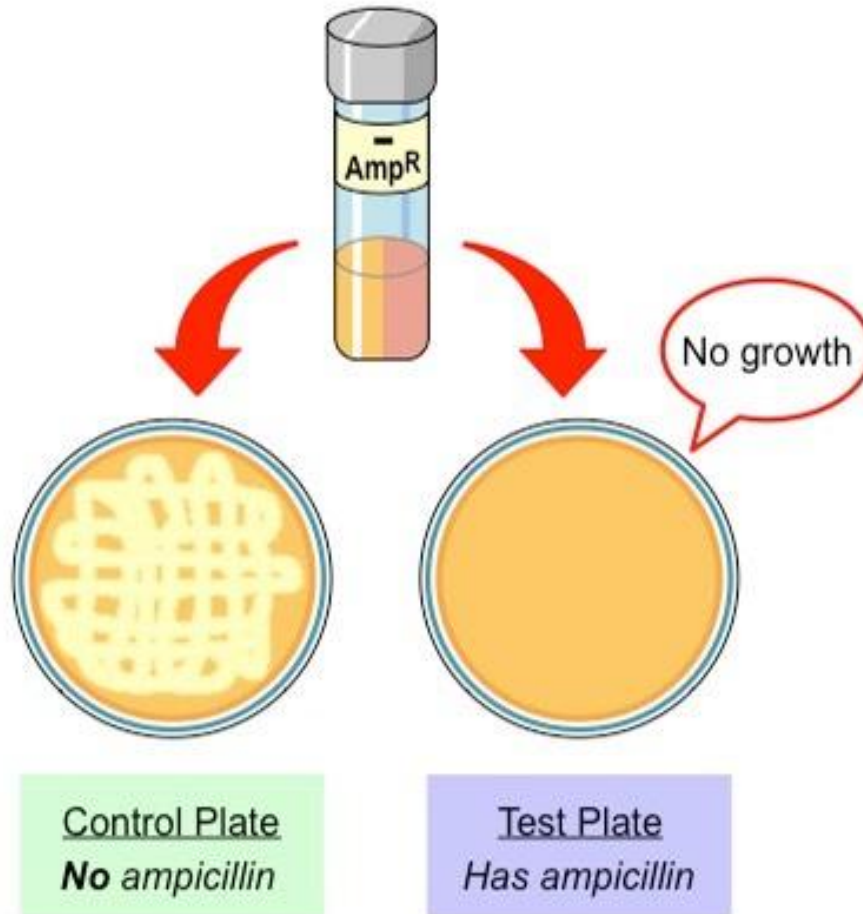


Promoter controls the binding of RNA polymerase to DNA, which transcribes DNA into mRNA that is then translated into a protein.

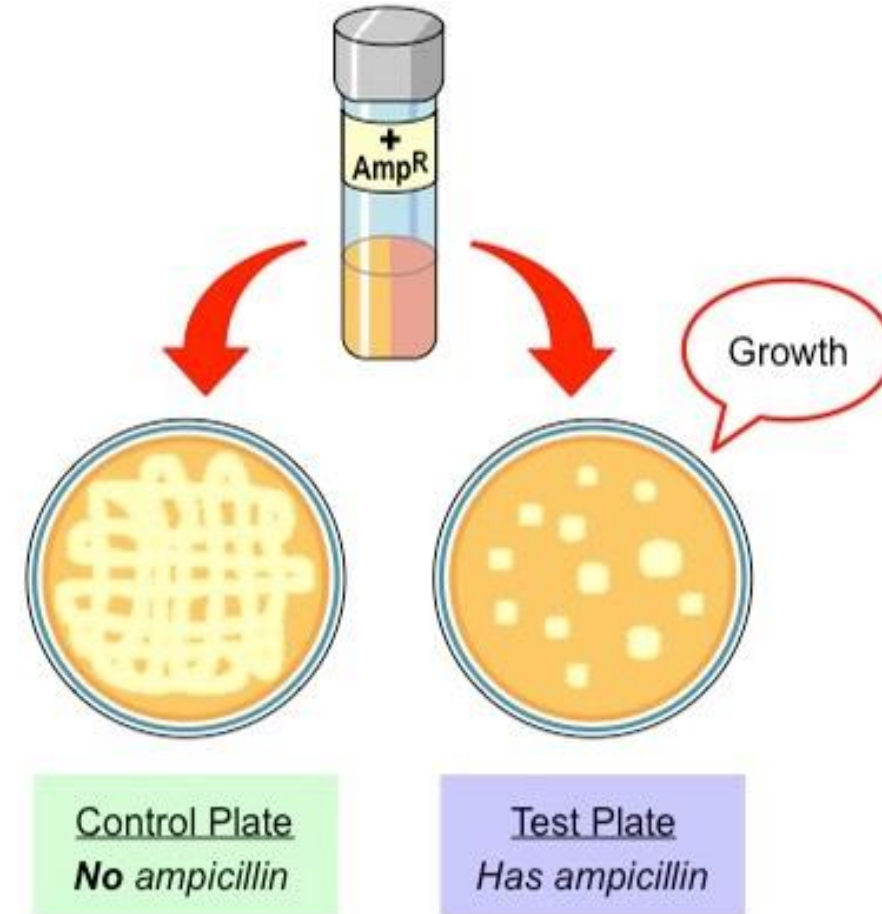
Selectable marker gene is a gene that gives a cell the ability to grow in the presence of toxic substances, such as antibiotics or herbicides, that would normally inhibit or kill the cell.

Isolation of Transgenic Cells via Antibiotic Selection

Control Condition
No Amp^R plasmids



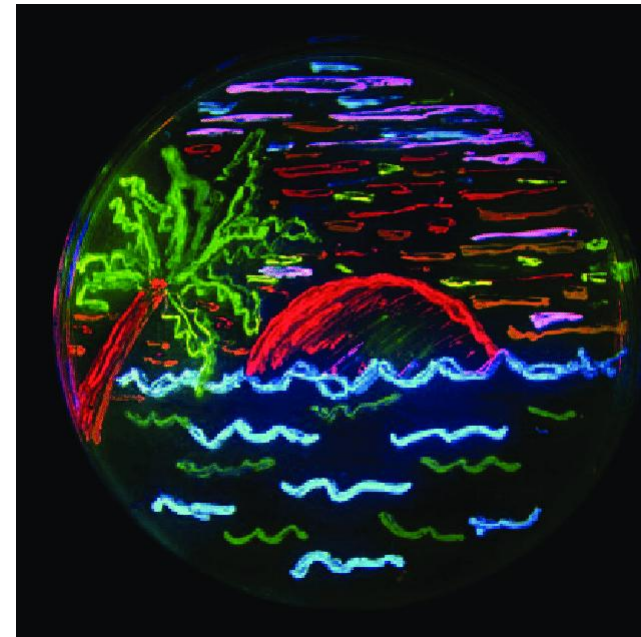
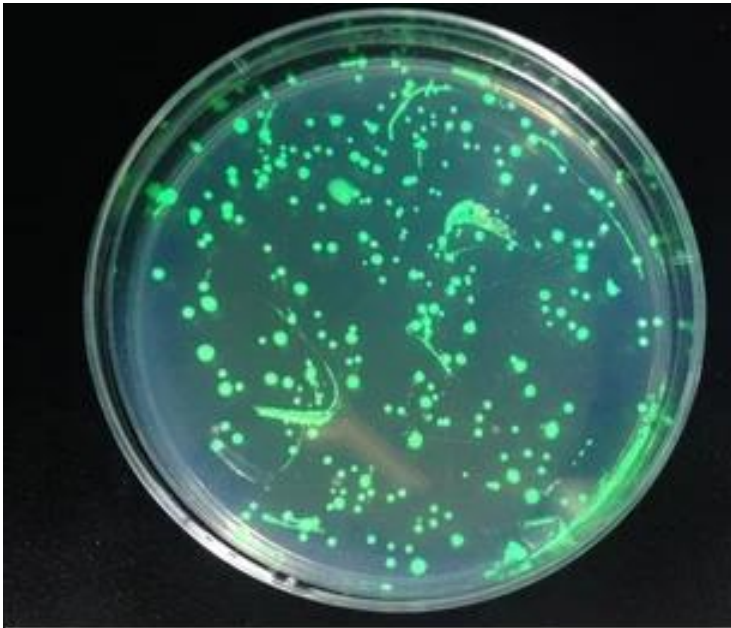
Experimental Condition
Amp^R plasmids added



Workflow of rDNA technology

Obtaining Foreign Gene Product

The recombinant DNA multiplies in the host and is expressed as a protein, under optimal conditions. This is now a recombinant protein. Small volumes of cell cultures will not yield a large amount of recombinant protein. Therefore, large-scale production is necessary to generate products that benefit humans. For this purpose, vessels called **bioreactors** are used.

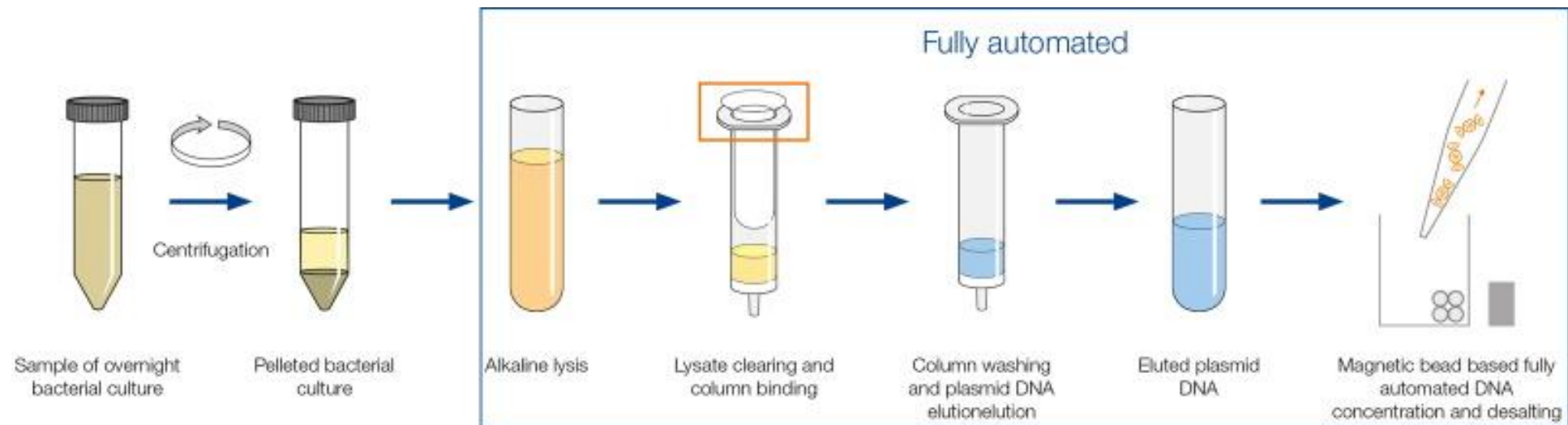


Harvesting your DNA

(Preparation sizes for plasmid DNA isolation)

Bacterial culture miniprep, midiprep, and maxiprep are methods for isolating plasmid DNA from bacterial cultures that differ in scale, yield, and the amount of starting bacterial culture:

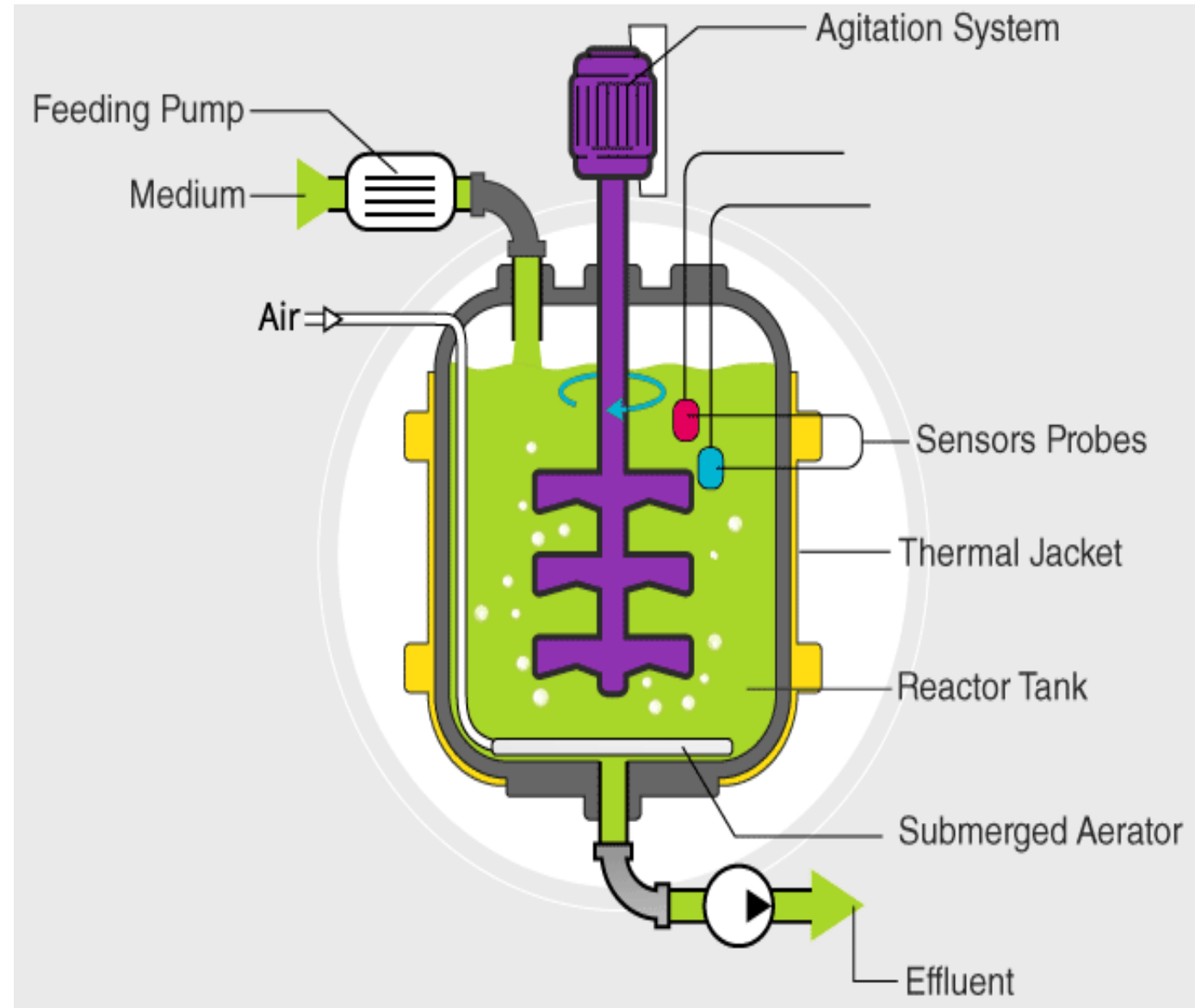
	Mini	Midi	Maxiprep	Megaprep	Gigaprep
Overnight bacterial culture volume	1–5 mL	25–100 mL	100–500 mL	500 mL–2.5 L	2.5–5 L
Approximate yield	Up to 40 μ g	Up to 400 μ g	up to 1.5 mg	Up to 5 mg	Up to 15 mg



Workflow of rDNA technology

Bioreactors are large containers with a continuous culture system, where the fresh medium is added from one side and used medium is taken out from another side. Bioreactors can process about 100-1000 litres of cell cultures.

A bioreactor provides optimum conditions (temperature, oxygen, pH, vitamins etc.) to biologically convert raw materials into specific proteins, enzymes etc.

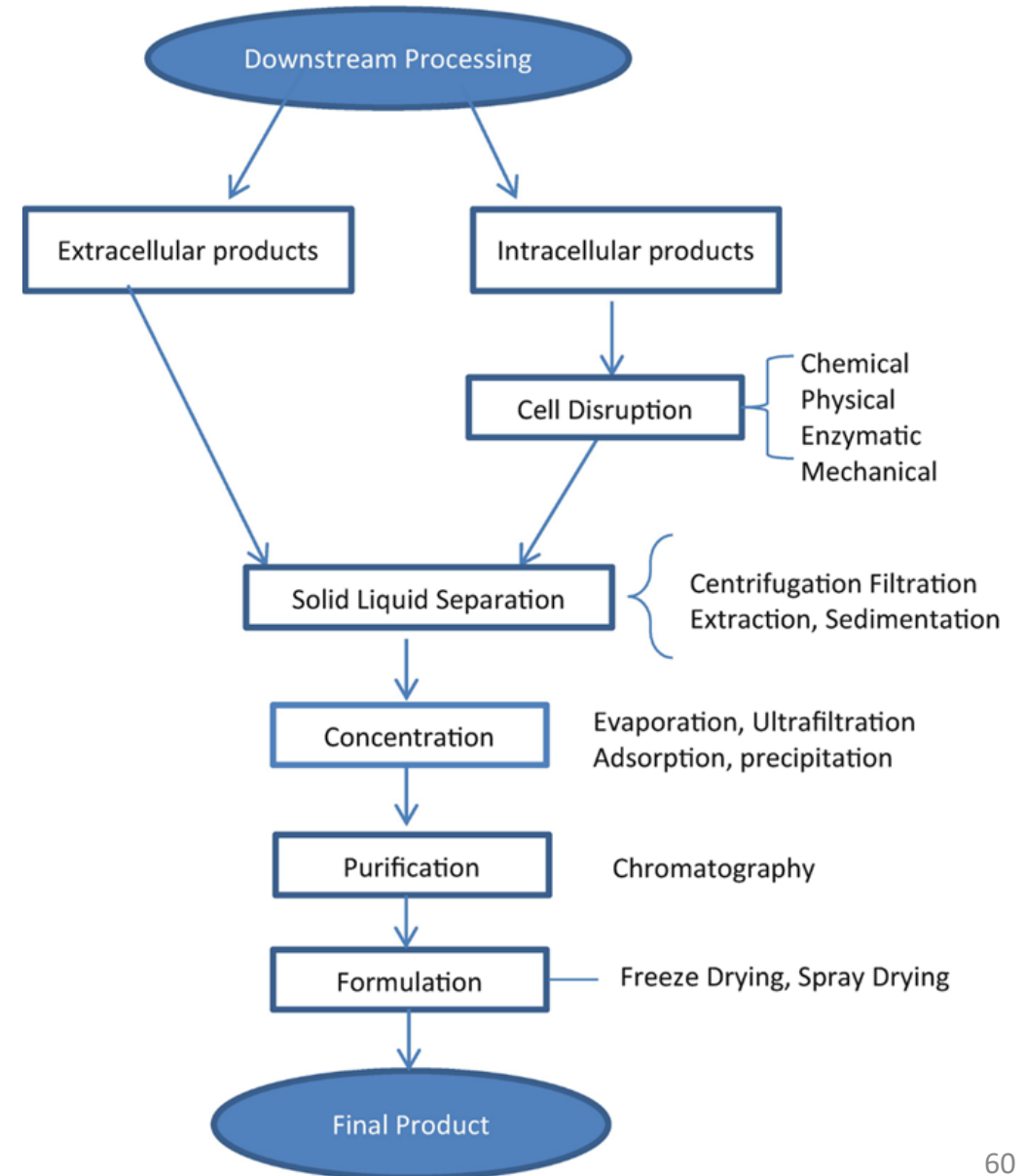


Downstream processing

Downstream Processing

Before the protein is marketed as a final product, it is subjected to downstream processing which includes:

- Separation and purification.
- Formulation with suitable preservatives.
- Clinical trials to test the efficacy and safety of the product.
- Quality control tests.



Applications of rDNA technology

Recombinant DNA (rDNA) technology has many applications, including:

Medicine: rDNA technology has been used to develop new drugs, vaccines, and pharmaceuticals. It can also be used to treat genetic disorders by repairing or replacing faulty genes. For example, rDNA technology has been used to produce synthetic human insulin and erythropoietin.

Agriculture: rDNA technology has been used to create genetically modified crops that are resistant to herbicides and pests. For example, Bt-cotton has been genetically modified to protect the plant from boll worms.

Food industry: rDNA technology can be used to improve the nutritional quality of food. For example, the nutritional value of soybeans has been improved by introducing a gene from the Brazil nut.

Bioremediation: rDNA technology can be used to clean up polluted soil and water by introducing genetically modified organisms (GMOs) into the site.

Diagnostics: rDNA technology can be used to diagnose infectious diseases and genetic diseases. For example, DNA fingerprinting can be used to determine paternity.

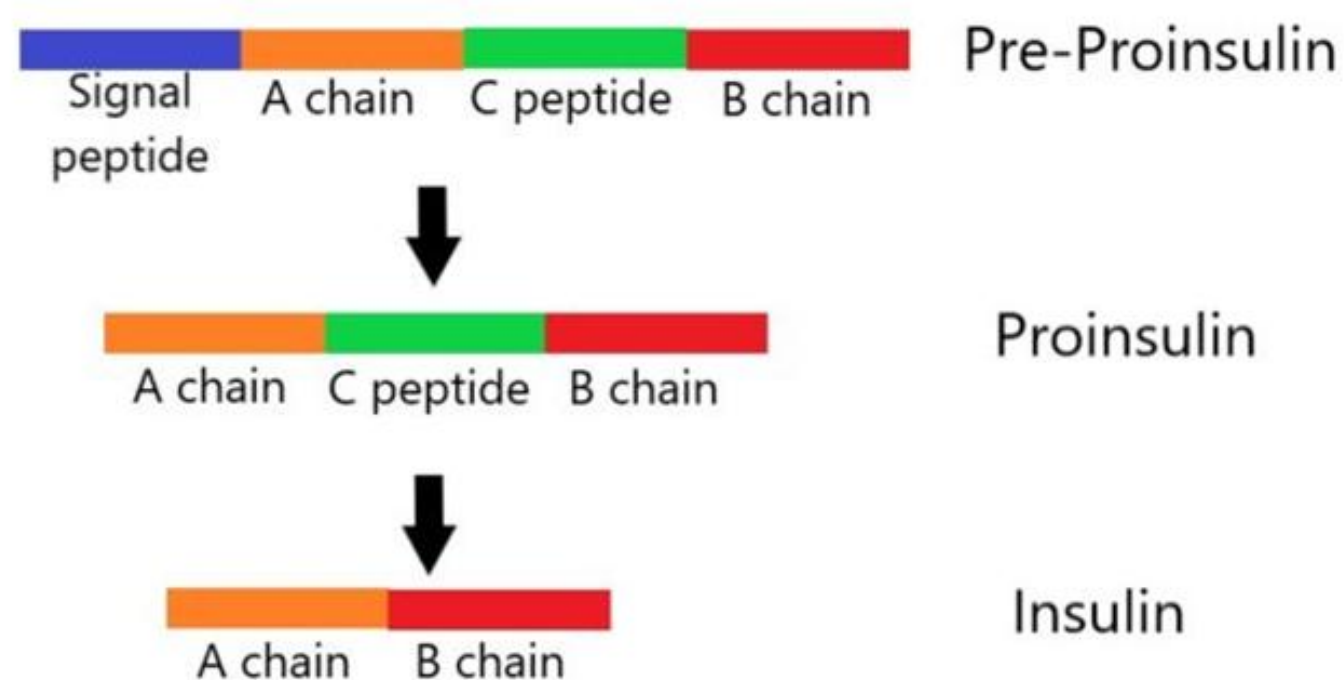
Immunotherapy:

rDNA technology can be used to develop immunotherapies, such as T-cell therapy, which uses the body's immune system to destroy cancer cells.

Biotechnology Applications In Medicine

Genetically Engineered Insulin:

- They prepared two DNA sequences that correspond to the A and B chains of human insulin. They then incorporated these sequences into plasmids of E. Coli to generate insulin chains.
- Further, they produced the chains separately, extracted and combined them by creating disulfide bonds to give rise to human insulin.

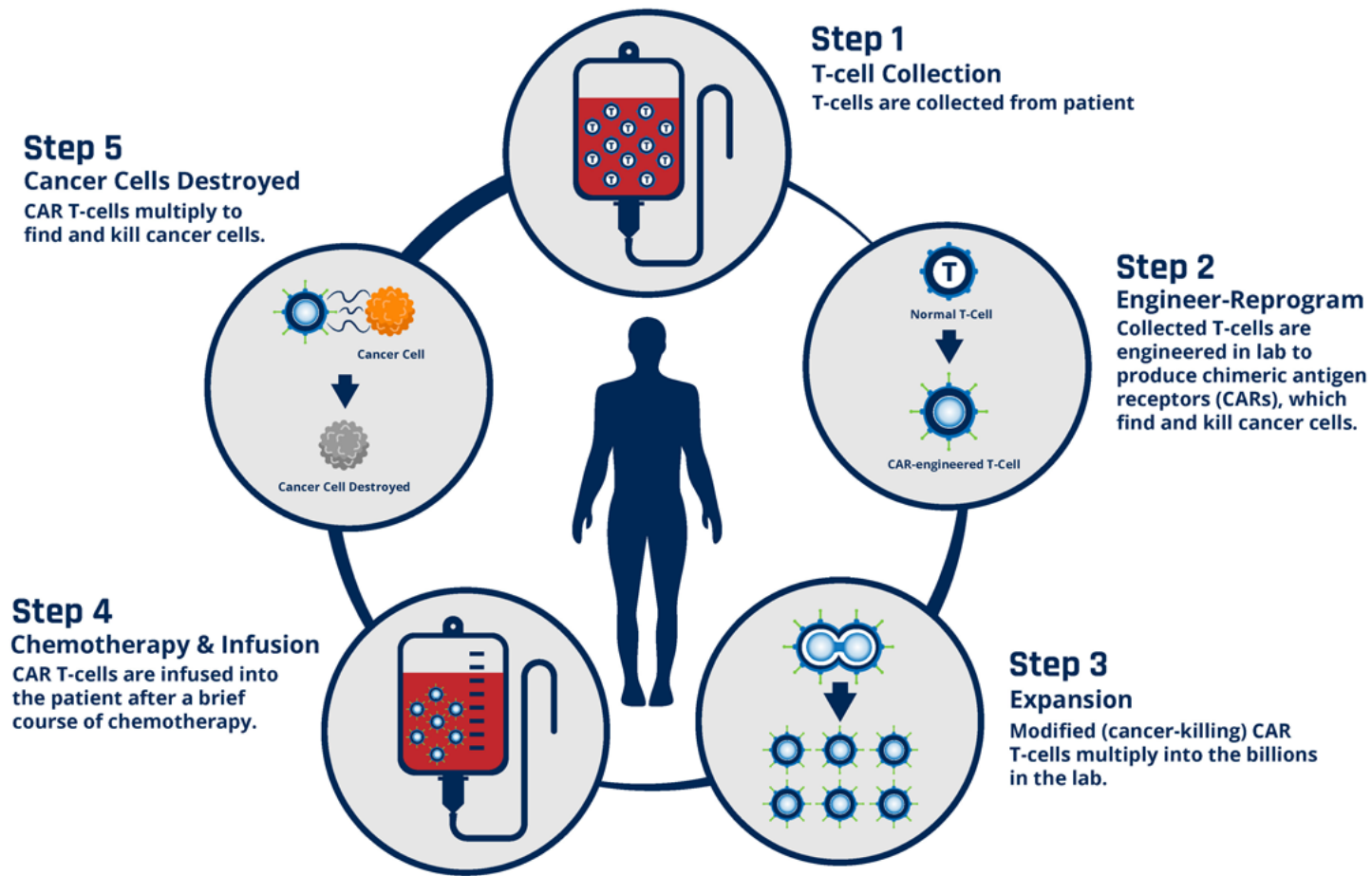


Biotechnology Applications In Medicine

Gene Therapy:

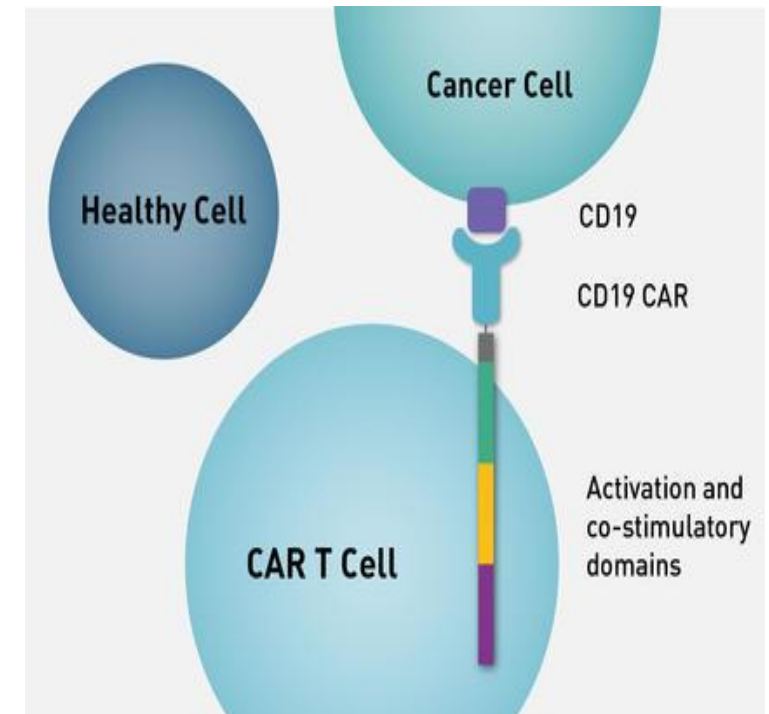
- If a child is born with a **genetic defect**, is there a way to correct that defect?
- Yes, there is, with gene therapy! Gene therapy is a biotechnology application involving a collection of methods that can correct a gene defect in a child or an embryo. It involves inserting a normal gene into the person's cells or tissues to compensate for the non-functional gene.
- In 1990, the first clinical gene therapy was applied to treat a 4-year old girl with a deficiency in the **enzyme adenosine deaminase** (ADA).
- This disorder is due to the lack of the gene for ADA, which is an enzyme important for the function of the immune system.
- **Bone marrow transplantation** helps cure this disorder in some cases. Enzyme replacement therapy, which involves injecting the patient with functional ADA, is also effective in some cases. However, both these procedures are not completely curative.

How Cell and Gene Therapy Works



<https://www.novartis.com/about/novartis-gene-therapies/what-cell-and-gene-therapy>

<https://www.asgct.org/education/more-resources/gene-and-cell-therapy-faqs>



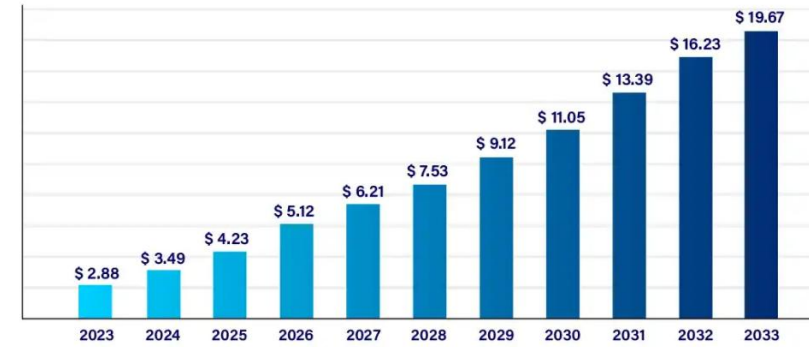
Global Cell and Gene Therapy Market

NOVA I ADVISOR Cell Therapy Market Size 2023 to 2033 (USD Billion)



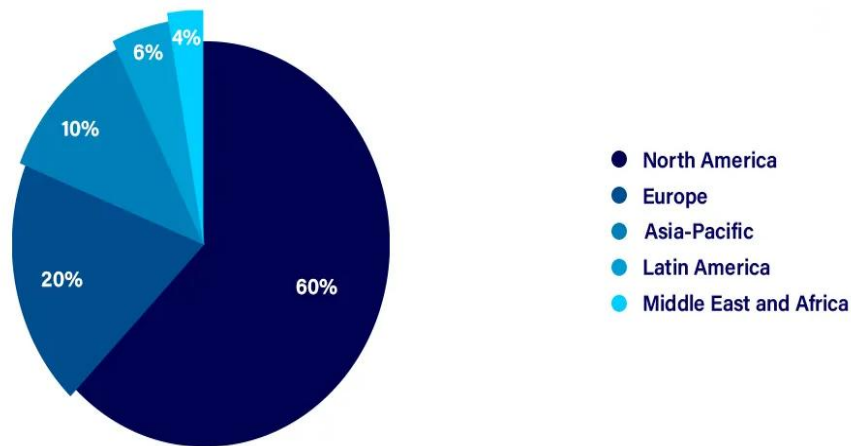
Source: www.novaoneadvisor.com

NOVA I ADVISOR U.S. Cell Therapy Market Size 2023 to 2033 (USD Billion)



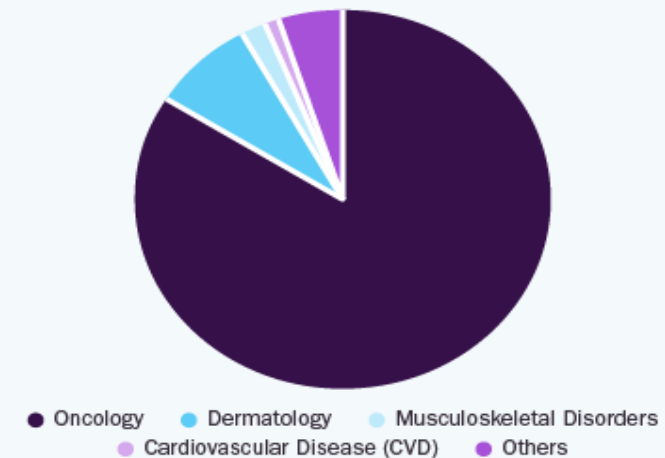
Source: www.novaoneadvisor.com

ISOR Cell Therapy Market Share, By Region, 2023 (%)



Global Cell Therapy Market

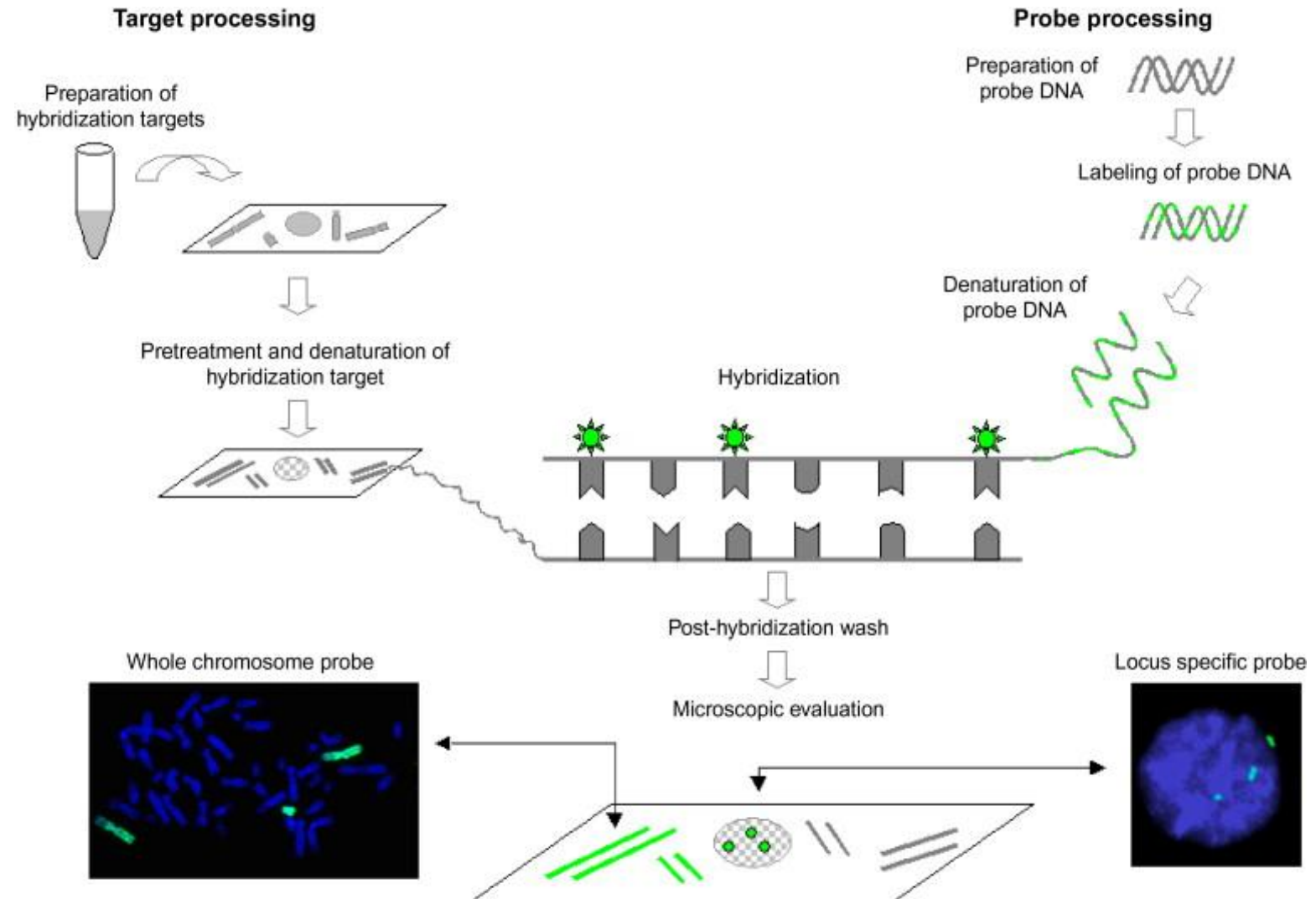
Share, by Therapeutic Area, 2023 (%)



Biotechnology Applications In Medicine

Molecular Diagnosis:

- In Situ Hybridisation: This technique involves tagging a single-stranded DNA or RNA with a radioactive molecule (probe). This then hybridizes with its complementary DNA in a clone of cells. On detection using autoradiography, the clone with the mutated gene will not appear on the photographic film because the probe is not complementary to the mutated gene.



Biotechnology Applications In Agriculture

- Application of Biotechnology in Agriculture involves scientific techniques such as Genetically Modified Organisms, Bt Cotton, Pest Resistant Plants. It helps in modifying plants, animals, and microorganisms and improve their agricultural productivity. Techniques like vaccines, tissue culture, genetic engineering are also used.
- **Genetically modified crops or GM crops are used in the following ways:**
 - ✓ They are more **tolerant to stresses** such as drought, cold, heat etc.
 - ✓ They are **pest-resistant** and therefore less dependent on chemical pesticides.
 - ✓ Genetically Modified crops help to **reduce post-harvest losses**.
 - ✓ They help to increase the mineral usage by plants, thereby preventing early exhaustion of **soil fertility**.
 - ✓ Genetically modified crops have enhanced nutritional value. Example – **Vitamin A enriched rice**.
 - ✓ Genetic modifications also help to create tailor-made plants to provide alternative resources to **industries, such as fuels, starches, and pharmaceuticals**.

Biotechnology Applications In Agriculture

Creating a Transgenic Crop

- A transgenic crop is genetically modified to be more useful to humans. The bacterium transfers the T-DNA (from the Ti plasmid) fragment with the desired gene into the host plant's nuclear genome.
- Transgenic crops have been created with a variety of different traits, such as yielding more food, tasting better, surviving drought, and resisting insect pests. Scientists have even created a transgenic purple tomato that contains a **cancer-fighting compound** and others that have high levels of **antioxidants**

Transgenic Purple Tomato. A purple tomato is genetically modified to contain a cancer-fighting compound. A gene for the compound was transferred into normal red tomatoes.



Biotechnology Applications In Agriculture

Bt Cotton:

This is a genetically modified version of cotton. 'Bt' stands for the microbe Bacillus thuringiensis.

This microbe produces an insecticidal **protein or toxin** that kills other insects such as tobacco budworm, flies, mosquitoes, beetles etc.



Golden Rice

Genetically modified variety of rice engineered to produce **beta-carotene**, a precursor to vitamin A, in the edible parts of the plant. It aims to combat vitamin A deficiency (VAD), a major public health concern in many developing countries. While it holds promise, Golden Rice has also faced controversy and hasn't yet been widely adopted.



Commonly Used Laboratory Animals

Mostly in Basic/simple research like Genetics, Gene Function etc.

Drosophila/(Fruit fly)



Zebrafish



Mice



Mostly in applied research like Vaccine, Hormone, Antibody etc.

Rat



Rabbit



Guinea pig



Monkey



Transgenic Animals and Ethical Issues

Normal Physiology and Development

- Transgenic animals allow us to study how genes affect the normal functions of the body and its development. In addition, they also help us understand how genes are regulated in the body.
- For example, by introducing genes from other species, that alter the formation of a particular factor and by studying its biological effects, we can gain knowledge about the biological role of the factor in our body.

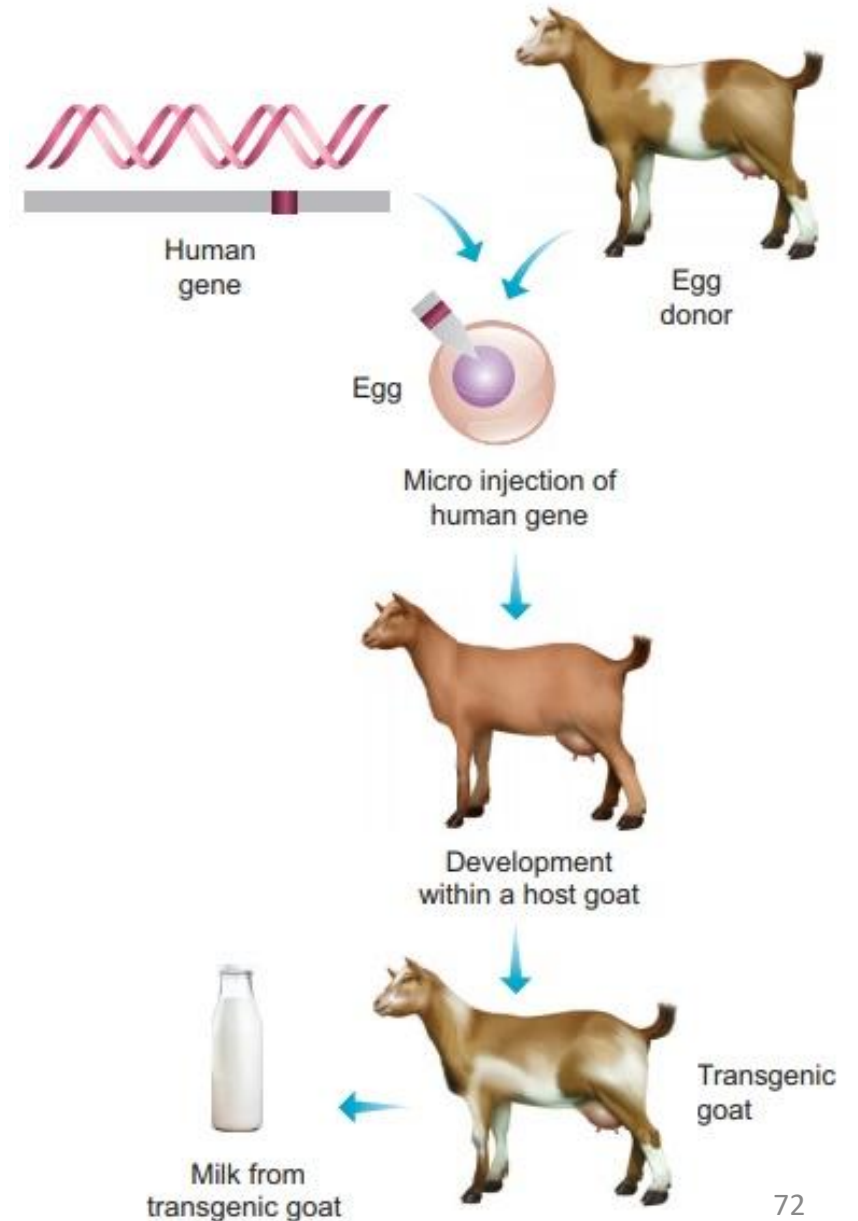
Study of Disease

- Transgenic animals also serve as models for human diseases and increase our understanding of how genes contribute to disease development. They also make it possible to investigate new treatment methods for diseases. Currently, transgenic models exist for many human diseases such as Alzheimer's, cancer, rheumatoid arthritis, cystic fibrosis etc.

Transgenic Animals and Ethical Issues

Biological Products

- Treatment of certain human diseases requires medicines that contain biological products, which are usually expensive to make. Transgenic animals can be used to make these biological products too. Incorporation of genes of a particular product into transgenics will result in the production of that biological product.
- An example is human α -1-antitrypsin which is used to treat emphysema. The most noteworthy example is Rosie – the first transgenic cow, generated in 1997. The milk from this cow contained human α -lactalbumin and had a higher nutritional value for human babies, compared to natural cow milk.



Transgenic Animals and Ethical Issues

Vaccine Safety

- Before using vaccines on humans, they are first tested for safety on transgenic mice. Transgenic mice are currently in use to test the safety of polio vaccines.
- If the test is successful and reliable, transgenic mice will replace the use of monkeys to test the safety of these vaccines.

Chemical Safety Testing

- The toxicity or safety of substances is also tested using transgenics. For this, transgenic animals are created with genes that make them more sensitive to the toxic substance under study. Subsequently, they are exposed to the toxic substance and the effects are studied. Using transgenics for toxicity testing gives results in a short time.

Transgenic Animals and Ethical Issues

Ethical Issues

- What do you think will happen if the genetic manipulation of living organisms by humans goes unregulated?!
- It will not only lead to exploitation of the organisms but will also have harmful impacts on our ecosystem. It is wrong at the moral and biological level!
- This is why we need ethical standards to regulate the human manipulation of organisms.
- In terms of genetically modified organisms (GMOs), the several Government has set up the Approval Committee and review committee which organization makes decisions regarding the validity of research involving GMOs and addresses the safety of GMOs introduced for public use.

Global and National Agencies Regulating GMO Food, Biosafety, and Transgenic Organisms

Region/Country	Agency/Organization	Responsibility
International	Cartagena Protocol on Biosafety (CBD)	Regulates transboundary movement of LMOs, promotes biosafety.
	Codex Alimentarius Commission (FAO/WHO)	Sets international food safety standards for GMOs.
United States	USDA (U.S. Department of Agriculture)	Regulates GM plants and crops for agricultural safety (e.g., pests).
	FDA (Food and Drug Administration)	Ensures safety of GMO food and animal feed.
	EPA (Environmental Protection Agency)	Regulates GMOs with pesticidal properties and environmental impact.
European Union	EFSA (European Food Safety Authority)	Conducts scientific risk assessments of GMOs.
	European Commission (DG SANTE, DG ENV)	Handles approvals, labelling, and traceability.
	National GMO Authorities	Implement EU regulations at the country level.
Bangladesh	National Committee on Biosafety (NCB)	Approves field trials and commercialization of GM crops.
	Bangladesh Agricultural Research Council (BARC)	Coordinates biosafety research and provides guidance.
	Biosafety Rules 2012	Legal framework governing GMOs and biosafety.
India	Genetic Engineering Appraisal Committee (GEAC)	Approves environmental release of GMOs.
India	Review Committee on Genetic Manipulation (RCGM)	Oversees research and trials involving GMOs.

<https://www.youtube.com/watch?v=iQsu3Kz9NYo>

PCR

<https://www.youtube.com/watch?v=ZDZUAleWX78>

Gel Electrophoresis

Please Watch

<https://www.youtube.com/watch?v=H9nSgxOoFlo>

https://www.youtube.com/watch?v=I_HwCeKr4Xg

<https://www.youtube.com/watch?v=MxV6FXPn6VU>

<https://www.susupport.com/knowledge/manufacturing-processes/single-use-technology/what-upstream-downstream-processing>